Index-Optimal Scheduling for Time-Varying Holding Costs

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Abstract

We study index policies for a two-class M/M/1 queue with identical service rates and increasing holding costs. We first introduce a discounted service penalty. The Threshold MDP compares allocations at fixed age thresholds, and the Index POMDP studies a change in one class's index with the other fixed. We then use the Penalty Game to ask how the service penalty at tied thresholds depends on the response times in the active queue. The proposed optimality argument and the vanishing-discount calculation have steps that remain open.

1 Introduction

We study a two-class M/M/1 queue with identical service rates. The age of a job is the time it has spent in the system, both in queue and in service. A class-i job of age t incurs holding cost at rate $c_i(t)$. We seek an index policy that minimizes total holding cost. Such a policy assigns a priority using an index function U_i for each class and serves the oldest job of the class with the higher priority.

For constant holding costs, the $c\mu$ rule is optimal. For increasing holding costs, the Whittle index of [1] assigns priority $\mu\mathbb{E}[c_i(t + X_i)]$ in the equal-service-rate case. Here $X_i \sim \text{Exp}(\mu - \lambda_i)$ is the response time when class i is served in isolation. In the same equal-rate case, if one class has constant holding cost, LookAhead uses this priority for the other class and is optimal among nondecreasing index policies under the assumptions of [2].

When both holding costs vary, we first consider the discounted problem with a service penalty ℓ. At fixed age thresholds, the Threshold MDP compares allocations among jobs that are eligible for service. In the Index POMDP, we fix one class's index and compare the priorities the other class can assign, including priorities below the service penalty. The proposed optimality argument requires the resulting pair of indices to be jointly optimal at each penalty.

In the Penalty Game we fix the service ordering and vary the thresholds at which the classes become eligible. Let τ_i be the threshold for class i, and let T_i be the time from eligibility to departure in the active queue, which contains only eligible jobs. The proposed vanishing-discount calculation gives

$$\ell = \mu\mathbb{E}[c_i(\tau_i + T_i)].$$

Here T_i depends on how the two classes share the server at those thresholds. The derivation requires showing that the optimal thresholds are tied under the given indices and justifying the limit in the threshold derivative.

Part I

Threshold MDP and Index POMDP

2 Problem Setup

We study a two-class M/M/1 queue with identical service rates, where each job incurs a time-varying holding cost for every unit of time it remains in the system. We serve jobs according to an *index policy*, a mapping from a each job's state to a real number priority. Our goal is to find the optimal index policy to minimize discounted total holding cost.

As in [1, Def 3.1], this problem can be translated to the following restless Multi-armed Bandit (RMAB) problem: where each arm represents a job class, whose state represents the age of the oldest job of the class. Our primary goal in this work is the following Problem 2.1: find the optimal *index* policy for this RMAB.

Problem 2.1 (IndexProblem(α))**.** We are given a 2-class M/M/1 queue where class i arrivals are of rate λ_i and have job size Exp(μ) (identically distributed for both classes). There are increasing instantaneous holding cost functions $c_i(t)$ where a class i job of age t (time spent in system so far) incurs holding cost at rate $c_i(t)$ dollars per second. For a given time-discount factor $\alpha > 0$, we define the following cost minimization problem.

Policy An index (by class) policy (U_1, U_2) such that the server always serves $\arg\max(U_1(t_1), U_2(t_2))$ when t_i is the age of the oldest class i job in the system. The server idles when there are no jobs in the system. Note, $U_i : \mathbb{R} \to \mathbb{R}^{\geq 0}$ maps oldest age to non-negative priority where $U_i(t) = 0$ for $t < 0$. In states where $U_1(t_1) = U_2(t_2)$, the server may choose to serve either job class.

Instantaneous Cost Suppose the age of the oldest class i jobs is t_i, the system accrues instantaneous holding cost

$$\alpha \,\text{hold}(t_1, t_2) := \alpha(r_1(t_1) + r_2(t_2)), \qquad \text{where} \quad r_i(t) := c_i(t) + \mathbb{E}\left[\sum_{a_j \in \text{Pois}_{\lambda_i}[0,t]} c_i(a_j)\right]$$

where $\text{Pois}_\lambda[0, t]$ refers to the set of arrival times of a Poisson arrival process with rate λ in the time interval $[0, t]$.

Objective Find the index policy which minimizes the total α-discounted holding cost accrued by the system, i.e. minimize

$$\int_0^\infty \alpha \,\text{hold}(S(t))e^{-\alpha t}dt$$

where $S(t)$ is the state (age of the oldest jobs at time t) starting from $S(0)$ being an idle system.

2.1 Penalty Problem

To facilitate analysis of Problem 2.1, we define a related problem with a service penalty ℓ. At time t, if the server serves any job (rather than idling), it incurs an additional service penalty at rate $\ell e^{-\alpha t}$. Note that the original IndexProblem(α) is obtained when $\ell = 0$. This is Problem 2.2.

Problem 2.2 (PenaltyIndexProblem(ℓ, α))**.** Given an M/M/1 system as described in Problem 2.1, time-discount factor $\alpha > 0$, and service penalty ℓ dollars per second, we define the following cost minimization problem.

Policy An index policy (U_1, U_2) such that the server serves $\arg\max(U_1(t_1), U_2(t_2))$ when t_i is the age of the oldest class i job in the system if $\max(U_1(t_1), U_2(t_2)) \geq \ell$ and idles otherwise.

Instantaneous Cost If the age of the oldest class i job is t_i, the system accrues instantaneous cost

$$\alpha \operatorname{hold}(t_1, t_2) + \ell \operatorname{serve}(t_1, t_2),$$

where

$$\operatorname{serve}(t_1, t_2) = \begin{cases} 1, & \text{if } U_1(t_1) \geq \ell \text{ or } U_2(t_2) \geq \ell, \\ 0, & \text{otherwise.} \end{cases}$$

Objective Find the index policy which minimizes the total α-discounted holding and ℓ service penalty costs accrued by the system, i.e. minimize

$$\int_0^\infty (\alpha \operatorname{hold}(S(t)) + \ell \operatorname{serve}(S(t)))e^{-\alpha t} dt,$$

starting from $S(0)$ being an idle system.

We abbreviate this problem as $\text{PIP}(\ell, \alpha)$.

2.2 Gittins Index

Definition 2.3 (Passive set)**.** Given $\ell, \alpha > 0$, let a pair of optimal index functions for $\text{PIP}(\ell, \alpha)$ be (U_1^*, U_2^*). We define the *passive set* for class i with respect to ℓ, α, $\mathcal{P}_i(\ell, \alpha)$ as the set of all ages at which the optimal index policy (U_1^*, U_2^*) prefers to idle over serving a class i job with age t. Namely,

$$\mathcal{P}_i(\ell, \alpha) = \{t \in \mathbb{R} : U_i^*(t) < \ell\}.$$

Conjecture 2.4 (Indexability)**.** The passive sets are increasing in ℓ. Namely, for every α, and $0 \leq \ell < \ell'$, we have

$$\mathcal{P}_i(\ell, \alpha) \subseteq \mathcal{P}_i(\ell', \alpha).$$

Definition 2.5 (Gittins Index)**.** Given $\alpha > 0$, we define the Gittins index of a class i job at age t to be the least service penalty at which the server prefers idling to serving the job in $\text{PIP}(\ell, \alpha)$. Namely,

$$\ell_i^*(t) = \inf\{\ell : t \in \mathcal{P}_i(\ell, \alpha)\}.$$

Conjecture 2.6 (Index Monotonicity)**.** The optimal index functions with respect to $\text{PenaltyIndexProblem}(\ell, \alpha)$ exist and are increasing (in age, not necessarily strictly).

2.3 Roadmap

Our goal is to show the Gittins index functions define an optimal policy to Problem 2.1.

In Section 3, we define a new $\text{ThresholdMDP}(\ell, \alpha, \tau_1, \tau_2)$ which builds on $\text{PIP}(\ell, \alpha)$: The policy is only allowed to serve class i jobs which are older than τ_i. We show that the optimal policy for some sets of age thresholds must behave "consistently" in Lemma 3.14.

In parallel, in Section 4, we view $\text{PIP}(\ell, \alpha)$ from the perspective of a one class, fixing the other class's index function. This yields a Partially observable MDP (POMDP) which we denote as IndexPOMDP. We derive Bellman criteria for a pair of index functions to be optimal in the classwise IndexPOMDPs, and thus form a pair of best responses for $\text{PIP}(\ell, \alpha)$.

In Section 5, we use both arguments in the proposed proof of Gittins optimality.

We use the conjectures outlined below. These conjectures are discussed formally in their respective sections.

Indexability Passive sets increase with the service penalty. See Conjecture 2.4.

Index Monotonicity Optimal index functions are increasing in age. See Conjecture 2.6.

Action Monotonocity Optimal actions in the threshold MDP are monotone in ages. See Conjecture 3.6.

Equilibrium Optimality A given set of Nash equilibria are globally optimal. See Conjecture 5.3.

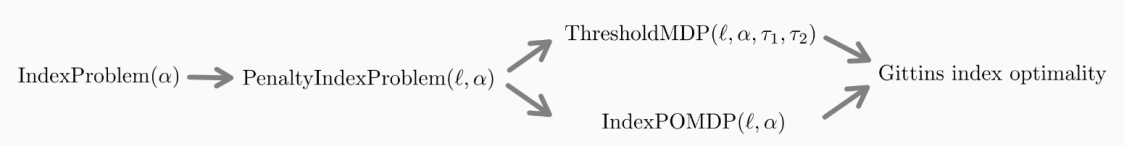


Figure 1: Roadmap of the proposed discounted proof.

2.4 Proposed result

Theorem 2.7 (Proposed)**.** *Given Conjectures 2.4, 2.6, 3.6 and 5.3, the Gittins index functions* (ℓ_1^*, ℓ_2^*) *are optimal index functions for* IndexProblem(α)*.*

3 Threshold MDP

Lemma 3.1. *Given Conjecture 2.6, for every* ℓ, α*,* $\mathcal{P}(\ell, \alpha) = \mathbb{R}$ *or there exists* $\tau_i^*(\ell, \alpha)$ *such that*

$$\mathcal{P}_i(\ell, \alpha) = (-\infty, \tau_i^*(\ell, \alpha))$$

Proof. U_i is increasing by Conjecture 2.6, thus either $\max_{t\in\mathbb{R}} U_i(t) \le \ell$ or there exists $\tau_i^*(\ell) = U_i^{-1}(\ell)$ such that the inverse image of $(-\infty, l)$ under U_i is $(-\infty, \tau_i^*(\ell))$. □

Lemma 3.2. *Given Conjectures 2.4 and 2.6, the Gittins index function* $\ell_i^*(\cdot)$ *is increasing.*

Proof. The threshold function $\tau_i^*(\cdot)$ defined in Lemma 3.1 is increasing since the passive sets are increasing by Conjecture 2.4. $\ell_i^*(\cdot)$ is the inverse function of $\tau_i^*(\cdot)$, thus it is also increasing. □

Problem 3.3 (ThresholdMDP($\ell, \alpha, \tau_1, \tau_2$))**.** Given an M/M/1 queue as described in Problem 2.1 as well as discount factor $\alpha > 0$, service penalty $\ell > 0$, age thresholds $\tau_1, \tau_2 \ge 0$, we define the following MDP

State (t_1, t_2), where $t_i \in \mathbb{R}$ represents the age of oldest class i job. If there are no class i jobs present in the system, $-t_i$ represents the time to next arrival.

Actions $\{\text{play}_1, \text{play}_2\}$. In every state, the policy chooses whether it would rather serve the oldest class 1 or class 2, if given a choice. The policy is described by function $A : \mathbb{R}^2 \to \{\text{play}_1, \text{play}_2\}$.

Execution If both $t_1 \ge \tau_1, t_2 \ge \tau_2$, the server serves the oldest class i job the policy action is play_i. Else if exactly $t_1 \ge \tau_1$ or $t_2 \ge \tau_2$, the server serves whichever class's oldest job is past its age threshold. If both $t_1 < \tau_1, t_2 < \tau_2$, then the server idles. The execution of this policy is described in Figure 2.

Objective Instantaneous Cost and Objective are the same as those described in Problem 2.2.

Lemma 3.4. *For* $\alpha > 0$ *and every finite* $\ell, \tau_1, \tau_2 \ge 0$*, under every policy* $A : \mathbb{R}^2 \to \{\text{play}_1, \text{play}_2\}$ *in* ThresholdMDP($\ell, \alpha, \tau_1, \tau_2$)*, the resulting Markov chain is ergodic.*

Proof. Every MDP policy A defines a work conserving scheduling policy for the active queue. Thus when $\lambda_1 + \lambda_2 < \mu$, we know the queueing system is stable and thus the Markov chain is ergodic. □

Proof status. The next argument uses ergodicity to reach an exact continuous state. That step is unproved. The later consistency argument also requires the replacement to be admissible and to preserve the discounted exit law. Even then, optimality of a restriction does not imply agreement of all optimal selectors.

Lemma 3.5 (Proposed)**.** A^* *is the optimal policy for* ThresholdMDP($\ell, \alpha, \tau_1, \tau_2$) *iff* $A^* = \arg\min_A V_A(S_0)$ *for some state* S_0*, where* $V_A(S_0)$ *is the* α*-discounted value function of* S_0 *under policy* A*.*

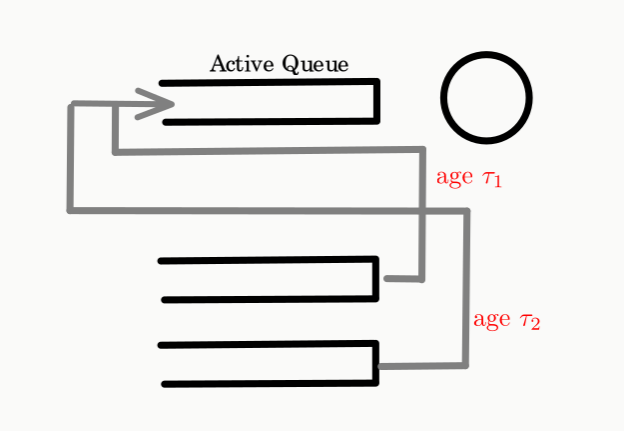


Figure 2: Execution of policy A in ThresholdMDP$(\ell, \alpha, \tau_1, \tau_2)$.

Proof. Suppose for the sake of contradiction there exists policy A' and state S_1 such that $V_{A'}(S_1) < V_{A^*}(S_1)$. Since the Markov chain under A^* is ergodic, the expected hitting time starting from state S_0 to reach state S_1 is finite. We form policy A'' which follows policy A^* until the first time we reach S_1, then follows A'. Then we have $V_{A''}(S_0) < V_{A^*}(S_0)$. This is a contradiction. □

Conjecture 3.6 (Action Monotonicity)**.** Let A^* be the optimal policy in ThresholdMDP$(\ell, \alpha, \tau_1, \tau_2)$.

$$(t_1, t_2) \xrightarrow{A^*} \text{play}_1 \quad \Longrightarrow \quad \forall t_1' \geq t_1, (t_1', t_2) \xrightarrow{A^*} \text{play}_1$$
$$(t_1, t_2) \xrightarrow{A^*} \text{play}_2 \quad \Longrightarrow \quad \forall t_2' \geq t_2, (t_1, t_2') \xrightarrow{A^*} \text{play}_2$$

Lemma 3.7. *Let A^* be the optimal policy in ThresholdMDP$(\ell, \alpha, \tau_1, \tau_2)$. Suppose in* TMDP$(\ell, \alpha, \tau_1, \tau_2)$*, for some $t_1\tau_1, t_2 \geq \tau_2$, A^* is indifferent to serving class* 1 *or class* 2*. Namely,*

$$(t_1, t_2) \xrightarrow{A^*} \text{play}_1 \qquad \textit{and} \qquad (t_1, t_2) \xrightarrow{A^*} \text{play}_2 \,.$$

Let $\mathcal{A} = \{(i, t) : i \in [2], t \geq t_i\}$. Given Conjecture 3.6, while there is at least one job with state in $\mathcal{A}$ in the system, A^* only serves jobs in $\mathcal{A}$. Namely, A^* is work conserving with respect to (t_1, t_2).

Proof. We can represent the system as in Figure 3, extending Figure 2. We mark class i jobs with age $\geq t_i$ as "important". The server serves jobs in the active or important queue. Now suppose for the sake of contradiction, there exists some state in which there exists some class 1 job with age $t_1' \geq t_1$ (or similarly class 2 job with age $t_2' \geq t_2$) and there server serves some job not in $\mathcal{A}$. Namely, the server serves some job in the active (non-important) queue while there exists a job in the important queue. The policy is only allowed to serve the oldest job of either class, so if it serves some job not in $\mathcal{A}$, it must serve the oldest class 2 with age $t_2'' < t_2$. Namely, the server serves a class 2 in the active queue when there is a class 1 job in the important queue. But we're given $(t_1, t_2) \xrightarrow{A^*} \text{play}_1$. By Conjecture 3.6, this means $(t_1', t_2) \xrightarrow{A^*} \text{play}_1$ and further $(t_1', t_2'') \xrightarrow{A^*} \text{play}_1$. Therefore the server must serve the class 1 job in the important queue optimally. We can similarly argue that the server must serve class 2 jobs in the important queue before serving any class 1 job in the active queue as well. This means the server is work conserving with respect to the important queue. □

Definition 3.8 (B, I, I_1, I_2)**.** We define B to be a random variable denoting the length of a 2-class M/M/1 busy period for the parameters described in Problem 2.1. Likewise, we define $I \sim \text{Exp}(\lambda)$ where $\lambda = \lambda_1 + \lambda_2$ to be the length of an idle period. We also define $I_i \sim \text{Exp}(\lambda_i)$ to be the inter-arrival times of class i jobs.

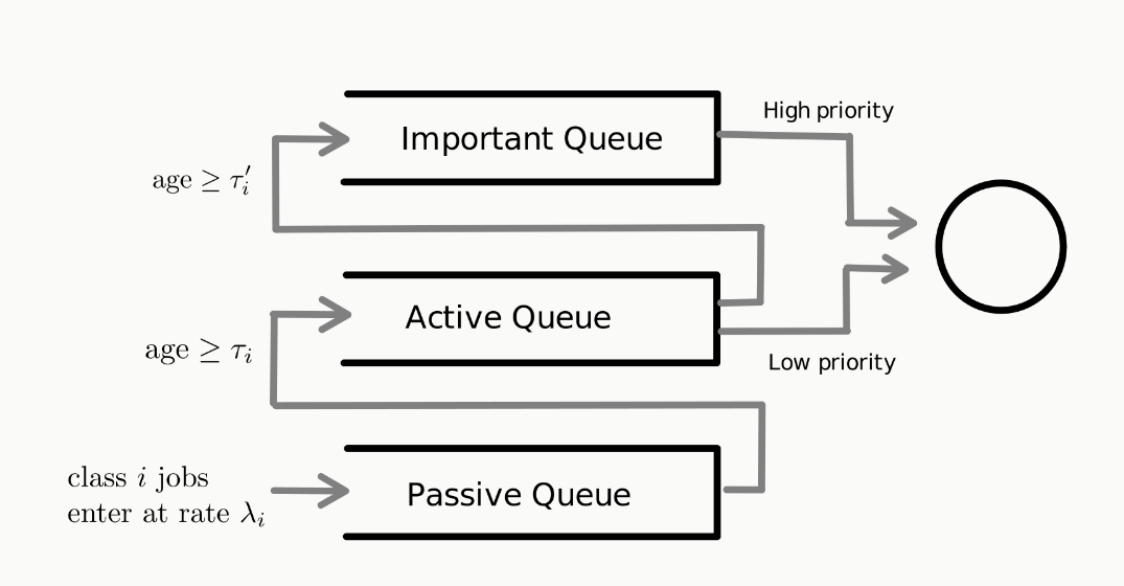


Figure 3: (τ_1', τ_2')-Important jobs in TMDP$(\ell, \alpha, \tau_1, \tau_2)$).

Definition 3.9 ($S_i(\tau_1, \tau_2)$)**.** Let $\tau_1, \tau_2 \geq 0$, $I_i \sim \text{Exp}(\lambda_i)$ for $i \in [2]$. We define the following (distributions of) states.

$$S_0(\tau_1, \tau_2) = (\tau_1 - I_1, \tau_2 - I_2)$$
$$S_{1\vee 2}(\tau_1, \tau_2) = \begin{cases} (\tau_1, \tau_2 - I_2) & \text{with probability } \frac{\lambda_1}{\lambda_1+\lambda_2} \\ (\tau_1 - I_1, \tau_2) & \text{with probability } \frac{\lambda_2}{\lambda_1+\lambda_2} \end{cases}$$

We also define the holding cost incurred during a busy period and an idle period by a policy as follows:

Definition 3.10 (holdB, holdI)**.** We define for $i \in \{1, 2, 1 \vee 2\}$, policy π which is work conserving with respect to (t_1, t_2)

$$\text{holdB}_i^\pi(t_1, t_2) = \mathbb{E}\left[\int_{t=0, S(0)=S_i(t_1,t_2)}^{\min T: S^\pi(T) \in (-\infty, t_1)\times(-\infty, t_2)} \alpha \, \text{hold}(S^\pi(t)) e^{-\alpha t} dt\right].$$

We also define

$$\text{holdI}(t_1, t_2) = \mathbb{E}\left[\int_{t=0}^{I} \alpha \sum_{i\in[2]} r_i(t_i - I_i + t) e^{-\alpha t} dt\right]$$

(Recall $r_i(\cdot)$ defined in Problem 2.1.) below

Lemma 3.11. *If $\alpha > 0$, A^* is the optimal policy for* ThresholdMDP$(\ell, \alpha, \tau_1, \tau_2)$ *iff it minimizes the discounted holding cost obtained in an active queue busy period. Namely, A^* is optimal iff*

$$A^* = \arg\min_A \int_{t=0}^{\min\{T: S(T) \in (\infty, \tau_1)\times(\infty, \tau_2)\}} \text{hold}(S^A(t)) e^{-\alpha t} \, dt$$

starting from state $S(0)$ distributed as $S_{1\vee 2}(\tau_1, \tau_2)$. Note that this is independent of ℓ.

Proof. For every policy A, the value of state $S_{1\vee 2}(\tau_1, \tau_2)$ i.e. the start of an active queue busy period can be expressed by unfolding the cost incurred during the first busy period and the first idle period.

$$V_A(S_{1\vee 2}(\tau_1, \tau_2)) = \text{holdB}_{1\vee 2}^A(\tau_1, \tau_2) + \mathbb{E}\left[e^{-\alpha B}\right] \text{holdI}(\tau_1, \tau_2) + \mathbb{E}[e^{-\alpha(B+I)}]\mathbb{E}[V_A(S_{1\vee 2}(\tau_1, \tau_2))]$$

Every MDP policy A represents a work-conserving scheduling policy with respect to the active queue, thus the length of the busy period, B is the same for all A. For this first busy period, we incur a holding cost $\text{hold}(S^A(t))$ where the state at time t depends on the policy A. We also incur service penalty at rate ℓ for the whole busy period. At the end of the busy period, we enter state $S_0(\tau_1, \tau_2)$, regardless of the policy A. The length of the idle period and the evolution of state are independent of the policy A. At the end of the idle period, we again enter state $S_{1\vee 2}(\tau_1, \tau_2)$, the start of a busy period. Solving, we obtain the following geometric series:

$$V_A(S_{1\vee 2}(\tau_1,\tau_2)) = \frac{\text{holdB}^A_{1\vee 2}(\tau_1,\tau_2) + \ell\left(\frac{1-\mathbb{E}[e^{-\alpha B}]}{\alpha}\right) + \mathbb{E}\left[e^{-\alpha B}\right]\text{holdI}(\tau_1,\tau_2)}{1-\mathbb{E}[e^{-\alpha(B+I)}]}. \tag{1}$$

Now, we know that policy A^* is optimal iff it minimizes $V_A(S_{1\vee 2}(\tau_1, \tau_2))$ by Lemma 3.5. Further, in Equation (1), we see that the only term in affected by A in $V_A(S_{1\vee 2}(\tau_1, \tau_2))$ is the holding cost incurred during an active queue busy period. Thus A^* is optimal iff it minimizes the holding cost incurred during an active queue busy period. □

Definition 3.12. We refer to $A^*(t_1, t_2)$ as the optimal policy in ThresholdMDP(α, τ_1, τ_2). Note that this is independent of ℓ.

Definition 3.13 (Consistency)**.** Let $\alpha > 0$. Two policies $A, A' : \mathbb{R}^2 \to \{\text{play}_1, \text{play}_2\}$ are said to be consistent if for every $t_1 \geq \max(\tau_1, \tau_1'), t_2 \geq \max(\tau_2, \tau_2')$, we have

$$(t_1, t_2) \xrightarrow{A} \text{play}_i \qquad \Longleftrightarrow \qquad (t_1, t_2) \xrightarrow{A'} \text{play}_i\,.$$

That is, two policies (in different MDPs) are consistent if their action is the same in every state where the action has some effect in both MDPs. A set of policies is said to be consistent if every pair of policies in the set are consistent.

Lemma 3.14 (Proposed consistency)**.** *Suppose in $A^*(\tau_1, \tau_2)$, we have $\tau_1' \equiv \tau_2'$ for some $\tau_1' \geq \tau_1, \tau_2' \geq \tau_2$. Then $A^*(\tau_1, \tau_2)$, $A^*(\tau_1', \tau_2')$ are consistent.*

Proof. We consider the value of the state $S_{1\vee 2}(\tau_1', \tau_2')$ in TMDP(α, τ_1, τ_2). This is essentially the state where an arbitrary job enters an empty (τ_1', τ_2')-important queue in Figure 3. We know by Lemma 3.7, since $\tau_1' \equiv \tau_2'$ with respect to $A^*(\tau_1, \tau_2)$, $A^*(\tau_1, \tau_2)$ is work conserving with respect to this important queue. Therefore this state is the start of a "important" busy period. At the end of this busy period we will be in state $(\tau_1' - I_1, \tau_2' - I_2)$. (Note we may continue serving jobs from the active queue so the server may not be idle.) We can thus unfold the value of state $S_{1\vee 2}(\tau_1', \tau_2')$ in TMDP(α, τ_1, τ_2) as follows:

$$V_{A^*(\tau_1,\tau_2)}(S_{1\vee 2}(\tau_1',\tau_2')) = \text{holdB}^{A^*(\tau_1,\tau_2)}_{1\vee 2}(\tau_1',\tau_2') + \ell\left(\frac{1-\mathbb{E}[e^{-\alpha B}]}{\alpha}\right) + \mathbb{E}[e^{-\alpha B}]V_{A^*(\tau_1,\tau_2)}(S_0(\tau_1',\tau_2'))$$

where B is a random variable denoting the length of a 2-class busy period. Now we know by Lemma 3.11 applied to TMDP$(\alpha, \tau_1', \tau_2')$,

$$A^*(\tau_1',\tau_2') = \arg\min_A \int_{t=0,\ S(0)=S_{1\vee 2}(\tau_1',\tau_2')}^{\min\{T:S(T)\in\prod_{i\in[2]}(-\infty,\tau_i')\}} \text{hold}(S^A(t))e^{-\alpha t}dt$$

We also know that $A^*(\tau_1, \tau_2) = \arg\min_A V_{A^*(\tau_1,\tau_2)}(S_{1\vee 2}(\tau_1', \tau_2'))$ since $A^*(\tau_1, \tau_2)$ is optimal in TMDP(α, τ_1, τ_2). Suppose $A^*(\tau_1, \tau_2)$ did not also minimize the cost in the first (τ_1', τ_2')-important busy period, we construct a policy A' which follows $A^*(\tau_1', \tau_2')$ for this first busy period and copies $A^*(\tau_1, \tau_2)$ thereafter. Then we have

$$\begin{aligned} V_{A'}(S_{1\vee 2}(\tau_1',\tau_2')) &= \text{holdB}^{A^*(\tau_1',\tau_2')}_{1\vee 2}(\tau_1',\tau_2') + \ell\left(\frac{1-\mathbb{E}[e^{-\alpha B}]}{\alpha}\right) \\ &\quad + \mathbb{E}[e^{-\alpha B}]V_{A^*(\tau_1,\tau_2)}(S_0(\tau_1',\tau_2')) \qquad < \qquad V^{(\ell)}_{A^*(\tau_1,\tau_2)}(S_{1\vee 2}(\tau_1',\tau_2')). \end{aligned}$$

This yields a contradiction since A' obtains lower cost than the optimal policy $A^*(\tau_1, \tau_2)$. Therefore $A^*(\tau_1, \tau_2)$ also minimizes the cost in an (τ_1', τ_2')-important busy period. By Lemma 3.11, this implies that $A^*(\tau_1, \tau_2)$ is also optimal in TMDP$(\alpha, \tau_1', \tau_2')$. Thus $A^*(\tau_1, \tau_2), A^*(\tau_1', \tau_2')$ are consistent. □

4 Index POMDP

We reduce Problem 2.2 to a Partially Observable MDP from the perspective of one class, when we fix the index function of the other class.

Problem 4.1 (IndexPOMDP$(\ell, \alpha, i, U_{\neg i})$)**.** Given an M/M/1 system as described in Problem 2.1, service penalty $\ell \geq 0$, class i which we can control and class $\neg i$ whose index function $U_{\neg i}$ is given, we define the following Partially Observable MDP (POMDP):

Observable State $t_i \in \mathbb{R}$, the age of oldest class i job.

Complete State (t_1, t_2), age of the oldest jobs of class $1, 2$.

Action $u_i \in \mathbb{R}$, the priority index value which can be assigned to state t_i.

Execution The state-action map defines index function $U_i : \mathbb{R} \to \mathbb{R}$. The server executes index policy $(U_i, U_{\neg i})$. Namely, as specified in Problem 2.1, the server serves the oldest job of whichever class has highest priority index. It idles, if both classes' priority indices are less than ℓ.

Objective The instantaneous cost and objective functions are the same as those specified in Problem 2.2.

Bellman Criteria The optimal index function U_i^* satisfies for every $t_i \in \mathbb{R}$

$$U_i^*(t_i) = \lim_{\delta \to 0} \arg\min_{u \in \mathbb{R}^{\geq 0}} Q(t_i, \langle u, \delta, U_i^* \rangle) \tag{2}$$

where the Q-value $Q(t_i, \langle u, \delta, U_i \rangle)$ is defined as the the expected long-run discounted cost starting from observed state t_i, applying action u_i for the first δ time and following policy U_i for all times thereafter. To show that a policy U_i is optimal in IndexPOMDP$(\ell, \alpha, i, U_{\neg i})$, it suffices to show for all $t_i \in \mathbb{R}, u \geq 0$,

$$\lim_{\delta \to 0} Q(t_i, \langle u, \delta, U_i \rangle) - Q(t_i, \langle U_i(t_i), \delta, U_i \rangle) \geq 0.$$

Remark 4.2. The optimal index policy (U_1^*, U_2^*) in Problem 2.2 must provide optimal index functions in the class-wise IndexPOMDPs. Namely,

$$\begin{aligned} U_1^* &= \arg\min_{U_1} \text{IndexPOMDP}(\ell, \alpha, 1, U_2^*), \\ U_2^* &= \arg\min_{U_2} \text{IndexPOMDP}(\ell, \alpha, 2, U_1^*). \end{aligned} \tag{3}$$

Note that this is only a necessary condition. In general, a pair of (U_1^*, U_2^*) satisfying Equation (3) may form a pair of best responses, i.e. a Nash equilibrium; they need not form a globally optimal pair of index functions.

Proof status. The Q-value comparison below uses the law of the other class given the observations of the controlled class. Using only the current oldest age as a POMDP state would require a sufficiency argument under deviations. In addition, the unscaled short-time difference in (2) vanishes as δ goes to zero. The first-order difference must be retained. The following derivation is the proposed calculation of that difference and the resulting indifference conditions.

4.1 Q-value difference

For the remainder of this section, we assume $i = 1$ and the index function U_2 is provided without loss of generality.

Definition 4.3 (V)**.** We fix index functions U_1, U_2 and use $V(t_1, t_2)$ to refer to the infinite horizon discounted cost starting from state (t_1, t_2) as defined in the objective of PIP(ℓ, α) (See Problem 2.2) under policy (U_1, U_2).

Definition 4.4 (T_2)**.** Let T_2 be the distribution of class 2 oldest age given class 1's observations till the current state under IndexPOMDP$(\ell, \alpha, 1, U_2)$.

Since we want to reason about the minimum of the Q-value function for the Bellman criteria, we first prove a useful lemma about the difference between the Q-values at different action indices u and $u + \Delta$.

Lemma 4.5. *For $\ell \le u < u + \Delta$,*

$$Q(t_1, \langle u, \delta, U\rangle) - Q(t_1, \langle u + \Delta, \delta, U\rangle) = e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) \in (u, u + \Delta)] \\ \mathbb{E}[V(t_1, T_2 - I_2) - V(t_1 - I_1, T_2)|U_2(T_2) \in (u, u + \Delta)]$$

For $u < u + \Delta < \ell$,

$$Q(t_1, \langle u, \delta, U\rangle) - Q(t_1, \langle u + \Delta, \delta, U\rangle) = 0.$$

For $u < \ell < u + \Delta$,

$$\begin{aligned} &Q(t_1,\langle u + \Delta, \delta, U\rangle) - Q(t_1, \langle u, \delta, U\rangle) \\ &\quad = \Pr[U_2(T_2) \in (\ell, u + \Delta)]\ \mu\mathbb{E}[V(t_1, T_2 - I_2) - V(t_1 - I_1, T_2)|U_2(T_2) \in (\ell, u + \Delta)] \\ &\quad + \Pr[U_2(T_2) < \ell]\ (\mu\mathbb{E}[V(t_1, T_2) - V(t_1 - I_1, T_2)|U_2(T_2) < \ell] - \ell) \end{aligned}$$

Proof. For $u \ge l$, we examine the difference between $Q(t_1, \langle u, U_1\rangle)$ and $Q(t_1, \langle u + \Delta, U_1\rangle)$.

We can express the Q-value of the policy $\langle u, \delta, U\rangle$ which assigns index u to class 1 for the first δ time, then follows U using the following facts.

- Instantaneous cost $\alpha\mathbb{E}[\text{hold}(t_1, T_2)]$ is incurred regardless of the actions and transitions. Further, since $u \ge \ell$, we have at least 1 job in the active queue (the oldest class 1), thus the system serves some job and an additional service penalty $\ell\delta$ is also incurred.
- If $U_2(T_2) > u$, we serve the oldest class 2 job. If not, we serve the oldest class 1. The job we serve leaves with probability $\mu\delta$.

This yields

$$\begin{aligned} Q(t_1, \langle u, \delta, U_1\rangle) = (\alpha\mathbb{E}[\text{hold}(t_1, T_2)] + \ell)\delta + e^{-\alpha\delta}(1 - \mu\delta)V_U(t_1 + \delta) \\ + e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) > u]\mathbb{E}[V(t_1 + \delta, T_2 + \delta - I_2)|U_2(T_2) > u] \\ + e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) < u]]\mathbb{E}[V(t_1 + \delta - I_1, T_2 + \delta)|U_2(T_2) < u]. \end{aligned}$$

We can express a similar equation for action $u + \Delta$.

$$\begin{aligned} Q(t_1, \langle u, \delta, U_1\rangle) &= (\alpha\mathbb{E}[\text{hold}(t_1, T_2)] + \ell)\delta + e^{-\alpha\delta}(1 - \mu\delta)V_U(t_1 + \delta) \\ &\quad + e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) > u + \Delta]\mathbb{E}[V(t_1, T_2 - I_2)|U_2(T_2) > u + \Delta] \\ &\quad + e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) \in (u, u + \Delta)]\mathbb{E}[V(t_1, T_2 - I_2)|U_2(T_2) \in (u, u + \Delta)] \\ &\quad + e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) \le u]\mathbb{E}[V(t_1 - I_1, T_2)|U_2(T_2) < u] \\ Q(t_1, \langle u + \Delta, \delta, U_1\rangle) &= (\alpha\mathbb{E}[\text{hold}(t_1, T_2)] + \ell)\delta + e^{-\alpha\delta}(1 - \mu\delta)V_U(t_1 + \delta) \\ &\quad + e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) > u + \Delta]\mathbb{E}[V(t_1, T_2 - I_2)|U_2(T_2) > u + \Delta] \\ &\quad + e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) \in (u, u + \Delta)]\mathbb{E}[V(t_1 - I_1, T_2)|U_2(T_2) \in (u, u + \Delta)] \\ &\quad + e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) \le u]\mathbb{E}[V(t_1 - I_1, T_2)|U_2(T_2) < u] \end{aligned}$$

Note that only the third and fourth summands above differ between $Q(t_1, \langle u, \delta, U_1\rangle)$ and $Q(t_1, \langle u + \Delta, \delta, U_1\rangle)$. Thus

$$Q(t_1, \langle u, \delta, U\rangle) - Q(t_1, \langle u + \Delta, \delta, U\rangle) = e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) \in (u, u + \Delta)] \\ \mathbb{E}[V(t_1, T_2 - I_2) - V(t_1 - I_1, T_2)|U_2(T_2) \in (u, u + \Delta)].$$

□

Proposition 4.6. *For sufficiently small $\delta > 0$, the function*

$$u \mapsto Q(t_1, \langle u, \delta, U\rangle)$$

is continuous and differentiable in u.

Proof. If $u \geq \ell$, we have by Lemma 4.5,

$$\begin{aligned} Q(t_1, \langle u, \delta, U\rangle) - Q(t_1, \langle u+\Delta, \delta, U\rangle) &= e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) \in (u, u+\Delta)] \\ &\quad \mathbb{E}[V(t_1, T_2 - I_2) - V(t_1 - I_1, T_2) | U_2(T_2) \in (u, u+\Delta)] \\ \lim_{\Delta\to 0} \frac{Q(t_1, \langle u, \delta, U\rangle) - Q(t_1, \langle u+\Delta, \delta, U\rangle)}{\Delta} &= e^{-\alpha\delta}\mu\delta p_{U_2(T_2)}(u)\mathbb{E}\big[V(t_1, U_2^{-1}(u) - I_2) - V(t_1 - I_1, U_2^{-1}(u)\big] \end{aligned}$$

If $u < \ell$, clearly by Lemma 4.5, the derivative exists and is 0. Thus in both cases, the derivative exists. □

Remark 4.7. The general Bellman conditions Equation (2) requires us to find the argmin of the map $u \mapsto Q(t_1, \langle u, \delta, U_1^*\rangle)$. If u is a local minimum of this map, it should satisfy

$$\frac{dQ(t_1, \langle u, U_1\rangle)}{du}\Big|_{u=U_1(t_1)} = \lim_{\delta,\Delta\to 0} \frac{Q(t_1, \langle u, \delta, U\rangle) - Q(t_1, \langle u+\Delta, \delta, U\rangle)}{\delta\Delta}\Big|_{u=U_1(t_1)} = 0. \tag{4}$$

4.2 Necessary Bellman Criteria

Throughout this subsection, we fix ℓ, α and the optimal index policy for Problem 2.2, which we notate as (U_1, U_2). We derive necessary properties for (U_1, U_2) to be optimal by examining the Bellman conditions for IndexPOMDP$(\ell, \alpha, 1, U_2)$.

Lemma 4.8. *For t_1 such that $U_1(t_1) = u \geq \ell$, we must have*

$$\mathbb{E}[V(t_1, T_2 - I_2) - V(t_1 - I_1, T_2) | U_2(T_2) = u] = 0.$$

Proof. For $u \geq \ell$, we examine Condition 4. Taking $\delta, \Delta \to 0$ from Lemma 4.5, we have for $t_2 = U_2^{-1}(u)$[1]

$$\begin{aligned} Q(t_1, \langle u, \delta, U\rangle) - Q(t_1, \langle u+\Delta, \delta, U\rangle) &= e^{-\alpha\delta}\mu\delta \Pr[U_2(T_2) \in (u, u+\Delta)] \\ &\quad \mathbb{E}[V(t_1, T_2 - I_2) - V(t_1 - I_1, T_2) | U_2(T_2) \in (u, u+\Delta)] \\ \lim_{\delta,\Delta\to 0} \frac{Q(t_1, \langle u, \delta, U\rangle) - Q(t_1, \langle u+\Delta, \delta, U\rangle)}{\delta\Delta} &= \mu p_{U_2(T_2)}(u)\mathbb{E}[V(t_1, T_2 - I_2) - V(t_1 - I_1, T_2) | U_2(T_2) = u] \end{aligned}$$

Setting LHS= 0 from Equation 4, we have the desired result. □

Lemma 4.9. *If $U_1(t_1) = \ell$, we must have*

$$\ell = \mu\mathbb{E}[V(t_1, T_2) - V(t_1 - I_1, T_2) | U_2(T_2) < \ell].$$

Proof. At any t_1, we have $U_1(t_1) = \ell$ iff $Q(t_1, \langle \ell, U_1\rangle) = Q(t_1, \langle \ell - \Delta, U_1\rangle)$. Then,

$$\begin{aligned} Q(t_1, \langle \ell, \delta, U_1\rangle) &= \alpha\mathbb{E}[\text{hold}(t_1, T_2)]\delta + \Pr[U_2(T_2) > \ell]\Big(\ell\delta + e^{-\alpha\delta}\mu\delta\mathbb{E}[V(t_1+\delta, T_2+\delta-I_2) | U_2(T_2) > \ell]) \\ &\quad + e^{-\alpha\delta}(1-\mu\delta)\mathbb{E}[V(t_1+\delta, T_2+\delta) | U_2(T_2) > \ell]\Big) \\ &\quad + \Pr[U_2(T_2) < \ell]\Big(\ell\delta + e^{-\alpha\delta}(1-\mu\delta)\mathbb{E}[V(t_1+\delta, T_2+\delta) | U_2(T_2) < \ell] \\ &\quad + e^{-\alpha\delta}\mu\delta\mathbb{E}[V(t_1+\delta-I_1, T_2+\delta) | U_2(T_2) < \ell]\Big) \\ Q(t_1, \langle \ell - \Delta, \delta, U_1\rangle) &= \alpha\mathbb{E}[\text{hold}(t_1, T_2)]\delta + \Pr[U_2(T_2) > \ell]\Big(\ell\delta + e^{-\alpha\delta}\mu\delta\mathbb{E}[V(t_1+\delta, T_2+\delta-I_2) | U_2(T_2) > \ell]) \\ &\quad + e^{-\alpha\delta}(1-\mu\delta)\mathbb{E}[V(t_1+\delta, T_2+\delta) | U_2(T_2) > \ell]\Big) \\ &\quad + \Pr[U_2(T_2) < \ell]\Big(e^{-\alpha\delta}\mathbb{E}[V(t_1+\delta, T_2+\delta) | U_2(T_2) < \ell]\Big) \end{aligned}$$

[1] If U_2 is u on many values (an interval), the inverse must be drawn from the distribution $T_2 | U_2(T_2) = u$.

Only the terms where $U_2(T_2) < \ell$ and the class 1 job which is served departs differ between the two terms. Equating the Q-values and taking $\delta \to 0$, we obtain

$$\ell = \mu \mathbb{E}[V(t_1, T_2) - V(t_1 - I_1, T_2) | U_2(T_2) < \ell].$$

□

4.3 Evaluating Increasing Index Policies

Definition 4.10 (τ_1, τ_2)**.** Given increasing index functions U_1, U_2, let τ_1, τ_2 be the age thresholds $U_i(\tau_i) = \ell$. Note that we don't serve any class i jobs with age $< \tau_i$ under policy (U_1, U_2) in PIP(ℓ, α).

Lemma 4.11. *Suppose U_1, U_2 are both increasing and $U_1(\tau_1) = U_2(\tau_2) = u \geq l$. Let π be the index-by-class scheduling policy characterized by U_1, U_2 and let $\mathcal{A} = \{(i,t) : i \in [2], t \geq \tau_i\}$. While there is at least one job with state in $\mathcal{A}$ in the system, π only serves jobs in $\mathcal{A}$.*

Proof. We begin by noting that when the index functions U_1, U_2 are increasing, the index-by-class policy characterized by these index functions is equivalent to the index (by job) policy. Thus we examine the index policy characterized by U_1, U_2. Now since U_i are increasing for $i \in [2]$, we have $U_i(t_i) \geq u$ if $t_i \geq \tau_i$ and $U_i(t_i) \leq u$ if $t_i \leq \tau_i$. Thus $U(s) \geq u$ if $s \in \mathcal{A}$ and $U(s) \leq u$ if $s \notin \mathcal{A}$. Thus, if any job in $\mathcal{A}$ has higher (or equal) priority to any job not in $\mathcal{A}$. Thus we have that while there is at least one job in $\mathcal{A}$, π only serves jobs in $\mathcal{A}$. □

Definition 4.12 (I_1, I_2)**.** Let $I_i \sim \text{Exp}(\lambda_i)$ for $i \in [2]$.

Lemma 4.13. *Given increasing index functions U_1, U_2 in* IndexPOMDP$(\ell, \alpha, 1, U_2)$*, given the oldest job of class 1 is t_1, suppose the age of the oldest class 2 job is distributed T_2. Then we have*

$$[T_2 | U_2(T_2) < \ell] \sim \tau_2 - I_2.$$

Note that this is independent of t_1.

Proof. $U_2(T_2) < \ell$ iff there are no class 2 jobs in the active queue. In steady state (any time after there has been at least 1 class 2 arrival to the active queue), when there are no class 2 jobs in the active queue, there is an Exp(λ_2) arrival time for the next class 2 job entering the active queue. Thus its current age must be distributed $\tau_2 - \text{Exp}(\lambda_2)$. Note that this is independent of t_1. □

Lemma 4.14. *For any increasing index policy π, we can evaluate the value functions of states $S_{1\vee 2}(\tau_1, \tau_2)$, $S_0(\tau_1, \tau_2)$ as follows:*

$$V_\pi(S_{1\vee 2}(\tau_1, \tau_2)) = \frac{1}{1 - \mathbb{E}[e^{-\alpha(B+I)}]} \left(\text{holdB}^\pi_{1\vee 2}(\tau_1, \tau_2) + \ell \left(\frac{1 - e^{-\alpha B}}{\alpha} \right) + e^{-\alpha B} \, \text{holdI}(\tau_1, \tau_2) \right) \tag{5}$$

$$V_\pi(S_0(\tau_1, \tau_2)) = \frac{1}{1 - \mathbb{E}[e^{-\alpha(B+I)}]} \left(\text{holdI}(\tau_1, \tau_2) + e^{-\alpha I} \, \text{holdB}^\pi_{1\vee 2}(\tau_1, \tau_2) + \ell e^{-\alpha I} \left(\frac{1 - e^{-\alpha B}}{\alpha} \right) \right) \tag{6}$$

Proof. Starting from state $S_{1\vee 2}(\tau_1, \tau_2)$, by Lemma 4.11, we know we the optimal policy (for Problem 2.2) serves jobs for a 2-class busy period. During this busy period, we incur holding cost $\text{holdB}_{1\vee 2}(\tau_1, \tau_2)$ and service cost

$$\mathbb{E}\left[\ell \int_0^B e^{-\alpha t} dt \right] = \ell \left(\frac{1 - \mathbb{E}[e^{-\alpha B}]}{\alpha} \right).$$

At the end of this busy period, we enter state $S_0(\tau_1 - I_1, \tau_2 - I_2)$, the start of an idle period. During this idle period, we incur holding cost $\text{holdI}(\tau_1, \tau_2)$. Finally, we return to state $S_{1\vee 2}(\tau_1, \tau_2)$. Putting this together, we obtain

$$V_\pi(S_{1\vee 2}(\tau_1, \tau_2)) = \text{holdB}^\pi_{1\vee 2}(\tau_1, \tau_2) + \ell \left(\frac{1 - e^{-\alpha B}}{\alpha} \right) + e^{-\alpha B} \, \text{holdI}(\tau_1, \tau_2) + e^{-\alpha(B+I)} V_\pi(S_{1\vee 2}(\tau_1, \tau_2)).$$

Solving for $V_\pi(S_{1\vee 2}(\tau_1, \tau_2))$ we obtain Equation (5). Similarly expanding the value of $V_\pi(S_0(\tau_1, \tau_2))$ through the first idle and busy periods yields Equation (6). □

4.4 Necessary Bellman Criteria given Index Monotonicity

Lemma 4.15. *Given Conjecture 2.6, for every ℓ, α, $\mathcal{P}(\ell, \alpha) = \mathbb{R}$ or there exists $\tau_i^*(\ell, \alpha)$ such that*

$$\mathcal{P}_i(\ell, \alpha) = (-\infty, \tau_i^*(\ell, \alpha))$$

Proof. U_i is increasing by Conjecture 2.6, thus either $\max_{t\in\mathbb{R}} U_i(t) \leq \ell$ or there exists $\tau_i^*(\ell) = U_i^{-1}(\ell)$ such that the inverse image of $(-\infty, l)$ under U_i is $(-\infty, \tau_i^*(\ell))$. □

Definition 4.16 (τ_1, τ_2)**.** Given Conjecture 2.6, let τ_1, τ_2 be the age thresholds defined by Lemma 4.15 for PIP(ℓ, α).

Lemma 4.17. *Given Conjecture 2.6, the optimal index function must satisfy*

$$\ell = \mu\mathbb{E}[V(\tau_1, \tau_2 - I_2) - V(\tau_1 - I_1, \tau_2 - I_2)], \tag{7}$$

and for every t_1, t_2 such that $U_1(t_1) = U_2(t_2) \geq \ell$,

$$V(t_1, t_2 - I_2) = V(t_1 - I_1, t_2). \tag{8}$$

Proof. We obtain Equation (7) by combining Lemmas 4.9 and 4.13. Equation (8) follows from Lemma 4.8 and Conjecture 2.6, since $[T_2|U_2(T_2) = U_1(t_1)]$ is a constant variable with value $U_2^{-1}(U_1(t_1))$ when U_i are both increasing. □

Remark 4.18. Note, considering IndexPOMDP$(\ell, \alpha, 2, U_1)$ symmetrically, we will also have

$$\ell = \mu\mathbb{E}[V(\tau_1 - I_1, \tau_2) - V(\tau_1 - I_1, \tau_2 - I_2)], \tag{9}$$

Lemma 4.11 gives us a way to reason about busy periods under index policies. We use this to transform Lemma 4.17 into conditions in terms of the holding costs incurred during busy and idle periods. We first introduce several pieces of useful notation and some minor lemmas.

Lemma 4.19.

$$\frac{\mu}{\alpha}\frac{(1 - \mathbb{E}[e^{-\alpha I}])(1 - \mathbb{E}[e^{-\alpha B})]}{1 - \mathbb{E}[e^{-\alpha(I+B)}]} = \mathbb{E}[e^{-\alpha B}].$$

Proof. We have $\mathbb{E}[e^{-\alpha B}] = \frac{\mu}{\mu+\alpha+\lambda(1-\mathbb{E}[e^{-\alpha B}])}$. We can transform this to $\alpha + \lambda - \lambda\mathbb{E}[e^{-\alpha B}] = \mu\frac{1-\mathbb{E}[e^{-\alpha B}]}{\mathbb{E}[e^{-\alpha B}]}$. We also know $\mathbb{E}[e^{-\alpha I}] = \frac{\lambda}{\lambda+\alpha}$ since $I \sim \text{Exp}(\alpha)$. Thus,

$$\begin{aligned}
\frac{\mu}{\alpha}\frac{(1 - \mathbb{E}[e^{-\alpha I}])(1 - \mathbb{E}[e^{-\alpha B})]}{1 - \mathbb{E}[e^{-\alpha(I+B)}]} &= \frac{\mu}{\alpha}\frac{(1 - \frac{\lambda}{\lambda+\alpha})(1 - \mathbb{E}[e^{-\alpha B}])}{1 - \frac{\lambda}{\lambda+\alpha}\mathbb{E}[e^{-\alpha B}]} && (\text{Substituting } \mathbb{E}[e^{-\alpha I}] = \tfrac{\lambda}{\lambda+\alpha}) \\
&= \frac{\mu(1 - \mathbb{E}[e^{-\alpha B}])}{\alpha + \lambda - \lambda\mathbb{E}[e^{-\alpha B}]} && (\text{Substituting } \alpha + \lambda - \lambda\mathbb{E}[e^{-\alpha B}] = \mu\tfrac{1-\mathbb{E}[e^{-\alpha B}]}{\mathbb{E}[e^{-\alpha B}]}) \\
&= \mathbb{E}[e^{-\alpha B}].
\end{aligned}$$

□

Lemma 4.20. *Given Conjecture 2.6, the optimal index functions must satisfy*

$$\ell = \mu\left(\left(\frac{1}{1 - e^{-\alpha B}}\right)\text{holdB}_1(\tau_1, \tau_2) - \frac{1}{1 - e^{-\alpha(I+B)}}\left(\text{holdI}(\tau_1, \tau_2) + e^{-\alpha I}\,\text{holdB}_{1\vee 2}(\tau_1, \tau_2)\right)\right) \tag{10}$$

Proof. We substitute Lemma 4.14 in Equation (7).

$$\begin{aligned}
\ell &= \mu\mathbb{E}[V(S_1(\tau_1,\tau_2)) - V(S_0(\tau_1,\tau_2))] && \text{(by Equation (7))}\\
&= \frac{\mu}{1-e^{-\alpha(B+I)}}\left((1-e^{-\alpha I})\,\mathrm{holdB}^{\pi}_{1\vee 2}(\tau_1,\tau_2) - (1-e^{-\alpha B})\,\mathrm{holdI}(\tau_1,\tau_2) + \ell\frac{(1-e^{-\alpha I})(1-e^{-\alpha B})}{\alpha}\right) && \text{(by Lemma 4.14)}\\
&= \frac{\mu}{1-e^{-\alpha(B+I)}}\left(\frac{(1-e^{-\alpha I})\,\mathrm{holdB}^{\pi}_{1\vee 2}(\tau_1,\tau_2) - (1-e^{-\alpha B})\,\mathrm{holdI}(\tau_1,\tau_2)}{1-\frac{\mu}{\alpha}\frac{(1-e^{-\alpha I})(1-e^{-\alpha B})}{1-e^{-\alpha(I+B)}}}\right) && \text{(Solving for } \ell)\\
&= \frac{\mu}{1-e^{-\alpha(B+I)}}\left(\frac{(1-e^{-\alpha I})\,\mathrm{holdB}^{\pi}_{1\vee 2}(\tau_1,\tau_2) - (1-e^{-\alpha B})\,\mathrm{holdI}(\tau_1,\tau_2)}{1-e^{-\alpha B}}\right) && \text{(by Lemma 4.19)}
\end{aligned}$$

□

5 Gittins optimality

The Threshold MDP argument is used in the next lemma to identify the allocation above a tied pair of ages. The Index POMDP calculation is then used to choose the penalty at which the tied pair is proposed to be a pair of best responses.

Proof status. The proof stops at the proposed best-response step. Local indifference does not establish that either bid is optimal over all deviations, and the argument does not show that the price map attains every required ℓ. Even if the best-response step were established, the passage to a joint optimum is the separate conjecture 5.3. The two corollaries below are steps in that conditional argument.

Definition 5.1. Let U_1^*, U_2^* be optimal index functions for $\mathrm{IP}(\alpha)$.

Lemma 5.2 (Proposed)**.** *For every ℓ, α, there exist age thresholds τ_1, τ_2 with $U_1^*(\tau_1) = U_2^*(\tau_2)$ such that the index functions $U_i(t) = \frac{\ell}{U_i^*(\tau_i)} U_i^*(t)$ form a Nash equilibrium for* $\mathrm{PIP}(\ell, \alpha)$.

Proof. Let $\tau_1 \geq 0$, define $\tau_2(\tau_1) = U_2^{*-1}(U_1^*(\tau_1))$. We know the index functions (U_1^*, U_2^*) characterize the optimal policy in $\mathrm{ThresholdMDP}(\ell, \alpha, \tau_1, \tau_2(\tau_1))$, for any ℓ, by Lemma 3.14. Now consider the map

$$\begin{aligned}
f(\tau_1) = \mu\Bigg(&\left(\frac{1}{1-e^{-\alpha B}}\right)\mathrm{holdB}^{U_1^*,U_2^*}_{1\vee 2}(\tau_1,\tau_2(\tau_1))\\
&- \frac{1}{1-e^{-\alpha(I+B)}}\left(\mathrm{holdI}(\tau_1,\tau_2(\tau_1)) + e^{-\alpha I}\,\mathrm{holdB}^{U_1^*,U_2^*}_{1\vee 2}(\tau_1,\tau_2(\tau_1))\right)\Bigg).
\end{aligned}$$

Then the index functions $U_1(t) = \frac{\ell}{U_i^*(\tau_1)} U_1^*(t)$ and the index functions $U_2(t) = \frac{\ell}{U_1^*(\tau_1)} U_2^*(t)$ satisfy the Bellman criteria (Equation (2)) for $\mathrm{IndexPOMDP}(f(\tau_1), \alpha, 1, U_2)$ and $\mathrm{IndexPOMDP}(f(\tau_1), \alpha, 2, U_1)$ respectively. Thus these index functions form a Nash equilibrium for $\mathrm{PIP}(\ell, \alpha)$. □

Conjecture 5.3. The Nash equilibrium found by Lemma 5.2 is the global optimum for $\mathrm{PIP}(\ell, \alpha)$.

Corollary 5.4. *For every ℓ, α, we have*

$$U_1^*(\tau_1^*(\ell)) = U_2^*(\tau_2^*(\ell)).$$

Proof. Given Lemma 5.2 and Conjecture 5.3, $\tau_1^*(\ell), \tau_2^*(\ell)$ are given by some $\tau_1, \tau_2 : U_1^*(\tau_1) = U_2^*(\tau_2)$. □

Corollary 5.5. *The Gittins index policy, characterized by index functions $(i,t) \mapsto \ell_i^*(t)$ is optimal for* $\mathrm{IP}(\alpha)$.

Proof. By Corollary 5.4, the index functions $\ell_1^*(\cdot), \ell_2^*(\cdot)$ characterize a scheduling policy identical to (U_1^*, U_2^*). Consider any state (t_1, t_2) in ThresholdMDP$(0, \alpha, 0, 0)$. Suppose $\ell_1^*(t_1) \geq \ell_2^*(t_2)$ w.l.o.g. then the Gittins index policy elects to serve class 1. We also know by Corollary 5.4, $U_1^*(t_1) = U_2^*(\tau_2^*(\ell_1^*(t_1))$. By Conjecture 2.4, since $\ell_2^*(t_2)$, we have $t_2 < \tau_2^*(\ell_1^*(t_1))$. By Conjecture 2.6, this implies $U_2^*(t_2) \leq U_2^*(\tau_2^*(\ell_1^*(t_1))) = U_1^*(t_1)$. Therefore the index policy (U_1^*, U_2^*) also elects to serve class 1. □

Part II

The Penalty Game and the limit as the discount rate tends to zero

In the first part the penalty selects the eligible ages. We now fix the two indices used to decide which class to serve when both are eligible, and vary the eligibility strategies. The derivative calculation aims to express the common penalty as $\mu\mathbb{E}[c_i(\tau_i + T_i)]$, with T_i measured in the active queue at the proposed thresholds.

Proof status. The Penalty Game calculation assumes the tied thresholds and the restriction property sought in the first part. The original derivation below also varies a threshold while keeping the physical-age indices fixed. This can change the allocation at a given pair of active ages. The holding derivative therefore needs a justification for that variation, together with a uniform limit argument. The lemmas explicitly left unfinished in the proof calculations are part of this gap. The proposed converse from the common-level equation to optimality remains open.

6 The Penalty Game

Consider a two-class M/M/1 queue with identical service rates. Let the age of a job be the time it has spent in system so far (both in queue and in service). Each job may incur an age-varying holding cost for every unit of time it spends in system. We showed in [1, Def 3.1], that this problem of minimizing age-varying holding costs can be translated to a restless Multi-armed Bandit (RMAB) problem: Each arm represents a job class, whose state represents the age of the oldest job of the class.

Problem 6.1 (CostMDP(α))**.** We are given a 2-class M/M/1 queue where class i arrivals are of rate λ_i and have job size Exp(μ) (identically distributed for both classes). There are increasing instantaneous holding cost functions $c_i(t)$ where a class i job of age t (time spent in system so far) incurs holding cost at rate $c_i(t)$ dollars per second. For a given time-discount factor $\alpha > 0$, we define the following cost minimization problem.

State: (t_1, t_2), where the age of the oldest class i job is t_i.

Cost: Suppose the age of the oldest class i jobs is t_i. Then the system accrues instantaneous holding cost

$$\alpha \,\text{hold}(t_1, t_2) := \alpha(r_1(t_1) + r_2(t_2)), \qquad \text{where} \quad r_i(t) := c_i(t) + \mathbb{E}\left[\sum_{a_j \in \text{Pois}_{\lambda_i}[0,t]} c_i(a_j)\right]$$

where $\text{Pois}_\lambda[0, t]$ refers to the set of arrival times of a Poisson arrival process with rate λ in the time interval $[0, t]$. The expectation is over the ages of all the jobs in the system of type i. In essence, $r_i(t)$ is the expected total instantaneous cost incurred by class i jobs when the oldest job has age t and younger jobs had arrived according to a Poisson process with rate λ_i (i.e. with Exp(λ_i) inter-arrival times). See [1] for more details.

Action: serve_1 or serve_2. In state (t_1, t_2), the action serve_i serves the oldest job of class i, if $t_i \geq 0$.

Objective: Our goal is to minimize

$$\int_0^\infty \alpha \,\mathrm{hold}(S(t))e^{-\alpha t}dt$$

where $S(t)$ is the state (age of the oldest jobs at time t).

Conjecture 6.2 (Action Monotonicity)**.** Let A^* be the optimal policy in CostMDP(α).

$$\begin{aligned} A^*(t_1,t_2) = \mathrm{serve}_1 \qquad &\Longrightarrow \qquad \forall t_1' \geq t_1, A^*(t_1',t_2) = \mathrm{serve}_1 \\ A^*(t_1,t_2) = \mathrm{serve}_2 \qquad &\Longrightarrow \qquad \forall t_2' \geq t_2, A^*(t_1,t_2') = \mathrm{serve}_2 \end{aligned}$$

Definition 6.3 (Index Policy)**.** A policy $A : \mathbb{R}^2 \to \{\mathrm{serve}_1, \mathrm{serve}_2\}$ is said to be an index policy if it can be characterized by index functions $U_i : \mathbb{R} \to \mathbb{R}^{\geq 0}$ which map oldest age to non-negative priority, such that[2]

$$A(t_1,t_2) = \mathrm{serve}_1 \qquad \Longleftrightarrow \qquad U_1(t_1) \geq U_2(t_2).$$

An index policy is said to be *increasing* if it can be characterized by increasing (non-decreasing) index functions.

We first present a preliminary result, showing that the MDP-optimal policy in CostMDP(α) is an index policy.

Lemma 6.4. *Given Conjecture 6.2, there exists an optimal policy for* CostMDP(α) *that is an increasing index policy.*

Proof. Let A^* be the optimal policy for CostMDP(α). We define the following increasing index functions and show that they agree with policy A^*.

$$\begin{aligned} U_1(t_1) &= t_1, \\ U_2(t_2) &= \inf\{t_1 \geq 0 : A^*(t_1,t_2) = \mathrm{serve}_1\}. \end{aligned}$$

The index policy (U_1, U_2) characterizes A^*. Given (t_1, t_2) such that $A^*(t_1,t_2) = \mathrm{serve}_1$, then $t_1 \in \{t_1' \geq 0 : A^*(t_1',t_2) = \mathrm{serve}_1\}$. Therefore $U_2(t_2) \leq t_1 = U_1(t_1)$. Thus (U_1, U_2) does action serve_1. Similarly if $A^*(t_1,t_2) = \mathrm{serve}_2$, then $t_1 \notin \{t_1' \geq 0 : A^*(t_1',t_2) = \mathrm{serve}_1\}$. Therefore by Conjecture 6.2, every $t_1'' < t_1$ is not in $\{t_1' \geq 0 : A^*(t_1',t_2) = \mathrm{serve}_1\}$. Therefore $U_2(t_2) \geq t_1 = U_1(t_1)$. □

6.1 Penalty strategies and proposed result

We start by introducing a related problem – the Penalty Game – that we will use to characterize the tradeoff of serving or holding a job with state (class i, age t).

Problem 6.5 (PenaltyGame($\ell, \alpha, S_0, U_1, U_2$))**.** Given service penalty $\ell \geq 0$, discount factor $\alpha > 0$, initial state $S_0 \in \mathbb{R}^2$, and index functions $U_1, U_2 : \mathbb{R} \to \mathbb{R}$ which characterize an optimal policy for CostMDP(α). We define the following two-player common-interest game, where each class is a player. Player i has the following state and action spaces:

State: t_i, the age of the oldest class i job.

Action: play_i or idle_i. If the action idle_i is chosen, class i will not be served. If action play_i is chosen, class i may be served according to the rules outlined below.

The game has the following collective state and objective:

State: (t_1, t_2), where the age of the oldest class i job is t_i.

[2]Note that we can have $U_1(t_1) = U_2(t_2)$ in some state where A is indifferent to actions $\mathrm{serve}_1, \mathrm{serve}_2$.

Cost: In state (t_1, t_2), suppose class i chooses action a_i for $i \in [2]$. The system incurs holding cost, as in Problem 6.1. In addition, if either class chooses its play action, the system also incurs service penalty ℓ. Namely, we incur

$$\alpha \,\mathrm{hold}(t_1, t_2) + \ell \,\mathrm{serve}(a_1, a_2), \qquad \text{where} \quad \mathrm{serve}(a_1, a_2) := \mathbb{1}_{(a_1 = \mathrm{play}_1) \vee (a_2 = \mathrm{play}_2)}.$$

Execution: Given actions a_1, a_2, the server chooses to either serve the oldest job of some class, or serve neither. If neither class chooses to play, the server idles. If exactly one chooses to play, the server serves the oldest job of that class. If both classes choose to play, the server plays whichever class was assigned higher priority by index functions (U_1, U_2). Namely,

$$\begin{aligned}(\mathrm{idle}_1, \mathrm{idle}_2) &\to \mathrm{idle}\\ (\mathrm{play}_1, \mathrm{idle}_2) &\to \mathrm{serve}_1\\ (\mathrm{idle}_1, \mathrm{play}_2) &\to \mathrm{serve}_2\\ (\mathrm{play}_1, \mathrm{play}_2) &\to \mathrm{serve}_{\arg\max(U_1(t_1), U_2(t_2))}\end{aligned}$$

Objective: The objective of the game is to minimize the α-discounted total holding and service costs starting from the idle system. Namely, let S_0 be the state representing the idle system. We want to minimize over strategies $\{\pi_i : \mathbb{R} \to \{\mathrm{idle}_i, \mathrm{play}_i\}\}_{i \in [2]}$,

$$\eth(\ell, \alpha, S_0, \pi_1, \pi_2, U_1, U_2) = \mathrm{hold}(\alpha, S_0, \pi_1, \pi_2, U_1, U_2) + \ell \,\mathrm{serve}(\alpha, S_0, \pi_1, \pi_2).$$

Here, $\mathrm{hold}(\alpha, S_0, \pi_1, \pi_2, U_1, U_2)$ represents the α-discounted holding cost $\int_{t=0}^{\infty} \alpha \,\mathrm{hold}(S(t)) e^{-\alpha t} dt$ starting from state S_0 and following state transitions given by strategies π_1, π_2 and index functions U_1, U_2. Likewise, $\mathrm{serve}(\alpha, S_0, \pi_1, \pi_2)$ represents the α-discounted service time $\int_{t=0}^{\infty} \mathrm{serve}(S(t)) e^{-\alpha t} dt$.

Definition 6.6 (Threshold strategy)**.** Let $i \in [2], \tau_i \geq 0$. In PenaltyGame$(\ell, \alpha, S_0, U_1, U_2)$, the class i strategy $\pi_i : \mathbb{R} \to \mathrm{play}_i, \mathrm{idle}_i$, Thresh$(\tau_i)$ is defined as

$$\pi_i(t_i) = \mathrm{play}_i \qquad \iff \qquad t_i \geq \tau_i.$$

Since jobs only incur higher holding cost as they become older, while serving any job costs the same ℓ dollars per second, we should expect serving jobs to become more attractive as they become older. Namely, we should expect "threshold" strategies to be optimal in PenaltyGame(ℓ, α, U_1, U_2).

Conjecture 6.7 (Threshold Optimality)**.** Given PenaltyGame(ℓ, α, U_1, U_2) for index functions U_1, U_2 which are optimal in Problem 6.1, there must exist thresholds $\tau_1, \tau_2 \geq 0$ (which may be infinite) such that the optimal strategy for player i is Thresh(τ_i).

Theorem 6.8 (Proposed)**.** *Suppose Conjectures 6.2 and 6.7 hold and we are given index functions U_1, U_2 which are optimal in* CostMDP(0)*. Let τ_1, τ_2 such that $U_1(\tau_1) = U_2(\tau_2)$. Let $\tau_i + T_i^{U_1, U_2 \restriction \tau_1, \tau_2}$ be a random variable distributed as the stationary distribution of the response time of class i jobs under a policy which, when the age of the oldest class j job is t_j for $j \in [2]$,*

(i) does not serve class j if $t_j < \tau_j$, and

(ii) if both $t_1 \geq \tau_1$ and $t_2 \geq \tau_2$, then it serves $\arg\max(U_1(t_1), U_2(t_2))$. [3]

Then we must have

$$\mathbb{E}[c_1(\tau_1 + T_1^{U_1, U_2 \restriction \tau_1, \tau_2})] = \mathbb{E}[c_2(\tau_2 + T_2^{U_1, U_2 \restriction \tau_1, \tau_2})].$$

Conversely, suppose U_1, U_2 satisfy the above for every τ_1, τ_2 such that $U_1(\tau_1) = U_2(\tau_2)$, then U_1, U_2 are optimal in CostMDP(0)*.*

[3] as in Problem 6.5.

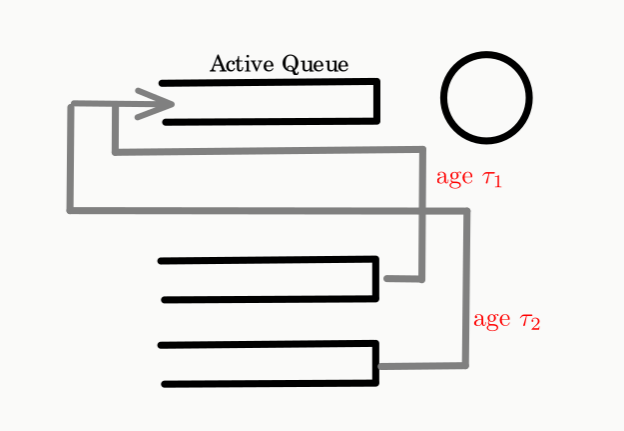


Figure 4: Execution of $\text{Thresh}(\tau_1), \text{Thresh}(\tau_2)$ in $\text{PenaltyGame}(\ell, \alpha, S_0, U_1, U_2)$.

Proposition 6.9 (Proposed)**.** *Consider Problem 6.1 where $c_2(\cdot)$ is a constant function valued at $c_2 \in \mathbb{R}$ and c_1 is strictly increasing. Suppose there exists τ_1^* such that $\mathbb{E}[c_1(\tau_1^* + T_1^{M/M/1})] = c_2$ where $T_1^{M/M/1} \sim \text{Exp}(\mu - \lambda_1)$. Then the optimal policy in Problem 6.1, $A : \mathbb{R}^2 \to \{1, 2\}$ is*

$$A(t_1, t_2) = 1 \qquad \textit{iff} \qquad t_1 \geq \tau_1^*.$$

Proof. For every $\tau_1 > \tau_1^*$, for every scheduling policy $\pi : \mathbb{R}^2 \to [2]$ which serves jobs FCFS within class, for every sample path ω, we have $T_1^{\pi}(\omega) \geq T_1^{M/M/1}(\omega)$. This is because every job can only be served by π after every older job of the same class has been served. Thus we have

$$\mathbb{E}[c_1(\tau_1 + T_1^{\pi})] \geq \mathbb{E}[c_1(\tau_1 + T_1^{M/M/1})] > \mathbb{E}[c_1(\tau_1^* + T_1^{M/M/1})] = c_2.$$

Therefore for all $\tau_1 > \tau_1^*$, there exists no τ_2 such that the optimal policy is indifferent (to serving 1 or 2) at (τ_1, τ_2). Thus in the region $(\tau_1^*, \infty) \times \mathbb{R}$, the optimal policy must strictly prefer 1 over 2 or 2 over 1.

Suppose the optimal policy strictly prefers 1 over 2 in the region $(\tau_1^*, \infty) \times \mathbb{R}$. Then at τ_1^*, we have $\mathbb{E}[c_1(\tau_1^* + T_1^{U_1, U_2 \mid \tau_1, \tau_2})] = \mathbb{E}[c_1(\tau_1^* + T_1^{M/M/1})] = c_2$, thus for every τ_2, the optimal policy is indifferent at (τ_1^*, τ_2). Thus by Conjecture 6.2, it is optimal to prefer 1 over 2 in $(\tau_1^*, \infty) \times \mathbb{R}$ and 2 over 1 in $(-\infty, \tau_1^*) \times \mathbb{R}$. Thus A is optimal. Else the optimal policy prefers 2 over 1 in the region $(\tau_1^*, \infty) \times \mathbb{R}$. Then Conjecture 6.2, the optimal policy must prefer 2 over 1 in all states. [4] □

6.2 Proof overview

To solve $\text{PenaltyGame}(\ell, \alpha, S_0, U_1, U_2)$ given Conjecture 6.7, the optimal thresholds τ_1^*, τ_2^* must satisfy

$$\text{For } i \in [2], \qquad \frac{\partial}{\partial \tau_i} \mathfrak{d}(\ell, \alpha, S_0, \tau_1, \tau_2, U_1, U_2)\Big|_{\tau_1^*, \tau_2^*} = 0. \tag{11}$$

Note that we can view the execution of $\text{Thresh}(\tau_1), \text{Thresh}(\tau_2)$ in $\text{PenaltyGame}(\ell, \alpha, S_0, U_1, U_2)$ as in Figure 4: class i jobs are held in an "idle" queue till age τ_i. They then join an active queue, where jobs are served according to $A : \mathbb{R}^2 \to \{\text{serve}_1, \text{serve}_2\}$ which is defined by U_1, U_2 truncated at τ_1, τ_2 respectively.

To facilitate calculating derivatives of hold and serve needed in Equation (1), we instead consider $\text{PenaltyGamePOMDP}_1\, \ell, \alpha, U_1,$ Here we fix player 2's strategy as π_2 and examine player 1's best response as a partially observable MDP (POMDP). Without loss of generality, we describe this POMDP from the perspective of class 1.

[4] TODO: why can't this happen?

Problem 6.10 (PenaltyGamePOMDP$_1(\ell, \alpha, U_1, U_2, \text{Thresh}(\tau_2))$)**.** Given PenaltyGame$(\ell, \alpha, S_0, U_1, U_2)$, where class 2 plays strategy Thresh(τ_2), class 1 is faced by the following POMDP. [5]

State: t_1, age of the oldest class 1 job.

Action: play$_1$ or idle$_1$.

Objective: Suppose T_2 is the distribution of class 2's oldest age given class 1 is in state t_1, the value of state t_1 under policy Thresh(τ_1) is $\eth(\ell, \alpha, (t_1, T_2), \tau_1, \tau_2, A^*)$. Thus our objective is to

$$\text{minimize}_{\tau_1} \eth(\ell, \alpha, (t_1, T_2), \tau_1, \tau_2, U_1, U_2)$$

Note that the optimality criteria in the POMDP are independent of initial state S_0. In particular, suppose Thresh(τ_1) is optimal in this POMDP, we must have

$$\lim_{\delta \to 0} \frac{\eth((\ell, \alpha, (\tau_1, T_2), \tau_1 + \delta, \tau_2, U_1, U_2)) - \eth((\ell, \alpha, (\tau_1, T_2), \tau_1, \tau_2, U_1, U_2))}{\delta} = 0.$$

That is, suppose we fix $S_1 = (\tau_1 + 1, T_2(\tau_1 + 1))$, where $T_2(t_1)$ is the stationary distribution of oldest class 2 job's age, given the oldest class 1 has age t_1. Then we must have

$$0 = \frac{\partial}{\partial \tau_1} \eth(\ell, \alpha, S_1, \tau_1, \tau_2, U_1, U_2) = \frac{\partial}{\partial \tau_1} \text{hold}(\alpha, S_1, \tau_1, \tau_2, U_1, U_2) + \ell \frac{\partial}{\partial \tau_1} \text{serve}(\alpha, S_1, \tau_1, \tau_2). \tag{12}$$

The following lemma is proved in Section 6.3.2.

Lemma 6.11 (Proposed)**.** *For any τ_1, τ_2 such that $U_1(\tau_1) = U_2(\tau_2)$[6], from any (t_1, t_2) such that $t_1 > \tau_1$ and $t_2 > \tau_2$ or $t_2 = \tau_2 - \text{Exp}(\lambda_2)$*

$$\lim_{\alpha \to 0} \frac{\partial}{\partial \tau_1} \text{hold}(\alpha, (t_1, t_2), \tau_1, \tau_2, U_1, U_2) = \lambda_1 \mathbb{E}[c_1(\tau_1 + T_1^{U_1, U_2 \restriction \tau_1, \tau_2})].$$

The following lemma is proved in Section 6.3.3.

Lemma 6.12 (Proposed)**.** *For any τ_1, τ_2 such that $U_1(\tau_1) = U_2(\tau_2)$[7], from any (t_1, t_2) such that $t_1 > \tau_1$ and $t_2 > \tau_2$ or $t_2 = \tau_2 - \text{Exp}(\lambda_2)$,*

$$\lim_{\alpha \to 0} \frac{\partial}{\partial \tau_1} \text{serve}(\alpha, (t_1, t_2), (\tau_1, \tau_2)) = -\rho_1.$$

Theorem 6.13 (Proposed)**.** *Suppose (U_1, U_2) is optimal in* CostMDP(0)*. For every τ_1, τ_2 such that $U_1(\tau_1) = U_2(\tau_2)$, we must have*

$$\mu \mathbb{E}[c_1(\tau_1 + T_1^{U_1, U_2 \restriction \tau_1, \tau_2})] = \mu \mathbb{E}[c_2(\tau_2 + T_2^{U_1, U_2 \restriction \tau_1, \tau_2})] = \ell$$

Proof. Let $S_1 = (\tau_1 + 1, T_2)$ be the state where class 1 has oldest age $\tau_1 + 1$ and T_2 is the stationary distribution of oldest class 2 job's age, given the oldest class 1 has age $\tau_1 + 1$. We must either have $T_2 > \tau_2$ with some probability or $T_2 \sim \tau_2 - \text{Exp}(\lambda_2)$. We take limit $\alpha \to 0$ on Equation (2) and use Lemmas 6.11 and 6.12.

$$\begin{aligned}
0 &= \lim_{\alpha \to 0} \frac{\partial}{\partial \tau_1} \eth(\ell, \alpha, S_1, \tau_1, \tau_2, U_1, U_2) \\
&= \lim_{\alpha \to 0} \frac{\partial}{\partial \tau_1} \text{hold}(\alpha, S_1, \tau_1, \tau_2, U_1, U_2) + \ell \frac{\partial}{\partial \tau_1} \text{serve}(\alpha, S_1, \tau_1, \tau_2) \\
&= \lambda_1 \mathbb{E}[c_1(\tau_1 + T_1^{U_1, U_2 \restriction \tau_1, \tau_2})] - \ell \rho_1
\end{aligned}$$

Thus we have $\ell = \mu \mathbb{E}[c_1(\tau_1 + T_1^{U_1, U_2 \restriction \tau_1, \tau_2})]$. Symmetrically, from class 2's perspective, we have $\ell = \mu \mathbb{E}[c_2(\tau_2 + T_2^{U_1, U_2 \restriction \tau_1, \tau_2})]$ □

[5] Note that this POMDP is independent of S_0.
[6] Note that U_1, U_2, τ_1, τ_2 are functions of α
[7] Note that U_1, U_2, τ_1, τ_2 are functions of α

6.3 Proof calculations

6.3.1 Tied thresholds and optimal allocation

Definition 6.14 (B, I, I_1, I_2)**.** We define B to be a random variable denoting the length of a 2-class M/M/1 busy period for the parameters described in Problem 6.1. Likewise, we define $I \sim \text{Exp}(\lambda)$ where $\lambda = \lambda_1 + \lambda_2$ to be the length of an idle period. We also define $I_i \sim \text{Exp}(\lambda_i)$ to be the inter-arrival times of class i jobs.

Lemma 6.15. *Given Conjectures 6.2 and 6.7, the optimal strategies* $\text{Thresh}(\tau_1), \text{Thresh}(\tau_2)$ *in* $\text{PenaltyGame}(\ell, \alpha, U_1, U_2)$ *must satisfy* $U_1(\tau_1) = U_2(\tau_2)$*.*

Proof to be written up.

Lemma 6.16. *Fix* $\alpha > 0$*. Suppose the nondecreasing index functions* U_1, U_2 *define a policy* π *that is optimal from every initial state in* $\text{CostMDP}(\alpha)$*, among admissible policies, including policies formed by switching at stopping times. Let* $\tau_1, \tau_2 \geq 0$ *be finite and satisfy* $U_1(\tau_1) = U_2(\tau_2) = u$*. If an eligible and an ineligible job have equal index, suppose* π *serves the eligible job. This condition holds automatically when the indices are strictly increasing.*

Assume $\lambda_1 + \lambda_2 < \mu$*, nonnegative holding costs, and finiteness of the discounted costs under consideration. Let* π^τ *be the restriction of* π *to class-*i *jobs of age at least* τ_i*. Then for every initial state* S_0*,*

$$\min_A \text{hold}(\alpha, S_0, \tau_1, \tau_2, A) = \text{hold}(\alpha, S_0, \tau_1, \tau_2, \pi^\tau),$$

where A *ranges over admissible nonidling allocations in the active queue. Thus* π^τ *is a common minimizer for all initial states.*

AI-written proof. We first compare allocations during one active busy period and then apply the comparison to successive busy periods.

One active busy period. Call a class-i job eligible when its age is at least τ_i. An eligible job has index at least u, whereas an ineligible job has index at most u. By the tie convention, π serves an eligible job whenever one is present. Hence π and π^τ agree until the active queue empties.

Start from a state S with at least one eligible job. Couple π and an arbitrary active allocation A using the same arrivals, initial jobs, and rate-μ potential-completion clock. Equal exponential service rates justify this clock under preemptive allocation. During an active busy period, neither policy serves ineligible jobs. The times at which jobs become eligible are therefore identical under the two policies. Each potential completion removes one eligible job under either policy, so their active counts agree until the common first emptying time B.

Their full configurations also agree at B. Every job that has become eligible by B has departed, and every remaining job is ineligible and has received no service in either system. In particular, the joint law of the exit state and B is independent of the allocation during this busy period. The same conclusion holds for the observable state used by the Cost MDP.

Let V_α^π be the value of π in $\text{CostMDP}(\alpha)$. The policy that follows A until B and then follows π is admissible. Optimality of π from S gives

$$\mathbb{E}_S\left[\int_0^B \alpha e^{-\alpha t} h_t^\pi \, dt + e^{-\alpha B} V_\alpha^\pi(S_B^\pi)\right] \leq \mathbb{E}_S\left[\int_0^B \alpha e^{-\alpha t} h_t^A \, dt + e^{-\alpha B} V_\alpha^\pi(S_B^A)\right],$$

where h_t denotes the total instantaneous holding cost. The continuation terms agree under the coupling. Subtracting them shows that π^τ minimizes the expected discounted holding cost up to B from S.

Successive busy periods. All Threshold MDP allocations idle when the active queue is empty. Consequently, the evolution during each intervening idle interval is allocation-independent. The coupling above also shows that the state at the end of each active busy period is allocation-independent.

Apply the first comparison conditionally at each active busy-period start. The Markov property and optimality of π from every state give the same inequality for each possible starting state. If the busy period starts at calendar time s, multiplying that inequality by $e^{-\alpha s}$ gives the comparison with discounting from time zero. Summing over the first n busy periods, and adding the identical idle-period costs, proves the desired inequality up to the end of the nth busy period. If the initial active queue is empty, the initial idle interval is also common to both policies.

Stability and finite thresholds imply that each active busy period ends almost surely. The arrival and potential-completion processes are nonexplosive, so the successive busy-period endpoints tend to infinity. Nonnegative holding costs allow passage to the infinite horizon by monotone convergence. Thus π^τ has no greater discounted holding cost than any admissible active allocation A, from every initial state S_0. □

6.3.2 Holding-cost derivative

Definition 6.17 (hold$(\alpha, S_0, \tau_1, \tau_2, A)$)**.** Given discount factor $\alpha > 0$, initial state $S_0 \in \mathbb{R}^2$, thresholds $\tau_1, \tau_2 \geq 0$, and active queue scheduling policy $A : \mathbb{R}^2 \to \{\text{serve}_1, \text{serve}_2\}$, we define

$$\text{hold}(\alpha, S_0, \tau_1, \tau_2, A) = \int_{t=0}^{\infty} \alpha \, \text{hold}(S(t)) e^{-\alpha t} dt$$

In the parametrization (τ_1, τ_2, A), the active queue sees jobs entering with age 0 (rather than τ_i). To be specific, the execution of policy (τ_1, τ_2, A) in state (t_1, t_2) is given as follows:

- If $t_1 < \tau_1, t_2 < \tau_2$, idle.
- If $t_1 \geq \tau_1, t_2 < \tau_2$, serve class 1.
- If $t_1 < \tau_1, t_2 \geq \tau_2$, serve class 2.
- If $t_1 \geq \tau_1, t_2 \geq \tau_2$, serve $A(t_1 - \tau_1, t_2 - \tau_2)$.

Definition 6.18 (Cumulative holding cost, $\bar{c}_i$)**.** For $i \in [2]$, we define $\bar{c}_i(t) = \int_0^t c_i(s) ds$ to be the cumulative holding cost incurred by a class i job when it has reached age t.

Lemma 6.19. *For any initial state $S_0 \in \mathbb{R}^2$,*

$$\lim_{\alpha \to 0} \text{hold}(\alpha, S_0, \tau_1, \tau_2, A) = \sum_{i \in [2]} \lambda_i \mathbb{E}[\bar{c}_i(\tau_i + T_i^A)]$$

Proof. As $\alpha \to 0$, hold$(\alpha, S_0, \tau_1, \tau_2, A)$ approaches the time average total holding cost incurred by the system. We can calculate this using the job average. Under policy (τ_1, τ_2, A), class i jobs have response time distribution $\tau_i + T_i^A$ where T_i^A is the response time of a class i job under the active queue scheduling policy A. Thus the class i job average holding cost is $\mathbb{E}[\bar{c}_i(\tau_i + T_i^A)]$. We have λ_i class i jobs per second. Thus by Little's Law, we have that the time average total holding cost is $\sum_{i \in [2]} \lambda_i \mathbb{E}[\bar{c}_i(\tau_i + T_i^A)]$. □

Definition 6.20 ($M_i^A(t)$)**.** Given 2 class M/M/1 scheduling policy $A : \mathbb{R}^2 \to \{\text{serve}_1, \text{serve}_2\}$, we define $M_i^A(t)$ to be age of the oldest class i job time t into an arbitrary busy period.

Lemma 6.21. *At τ_1, τ_2 such that $U_1(\tau_1) = U_2(\tau_2)$[8], we have*

$$\lim_{\alpha \to 0} \frac{\partial}{\partial \tau_1} \text{hold}(\alpha, (\tau_1 - I_1, \tau_2 - I_2), \tau_1, \tau_2, U_1, U_2) = \lambda_1 \mathbb{E}[c_1(\tau_1 + T_1^{U_1, U_2 \restriction \tau_1, \tau_2})]$$

Proof. Let A be an active queue scheduling policy. Under policy (τ_1, τ_2, A), from state $(\tau_1 - I_1, \tau_2 - I_2)$, it takes I_i time till the next class i arrival to the active queue. Thus from state $(\tau_1 - I_1, \tau_2 - I_2)$, we idle for

[8]Note that U_1, U_2, τ_1, τ_2 are functions of α

an idle period I, then serve jobs according to A for a busy period B. Then we return to the renewal event starting an active queue idle period. Thus we have

$$\text{hold}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2,A)=\frac{\mathbb{E}\Big[\int_0^{I+B}\alpha(r_1(\tau_1+M_1^A(t))+r_2(\tau_2+M_2^A(t)))e^{-\alpha t}dt\Big]}{1-\mathbb{E}[e^{-\alpha(I+B)}]}$$

$$\implies \frac{\partial}{\partial\tau_1}\text{hold}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2,A)=\frac{\mathbb{E}\Big[\int_0^{I+B}\alpha r_1'(\tau_1+M_1^A(t))e^{-\alpha t}dt\Big]}{1-\mathbb{E}[e^{-\alpha(I+B)}]}$$

To exchange the derivative and integral in the last line, note that

$$\mathbb{E}\left[\int_0^{I+B} r_1(\tau_1+h+M_1^A(t))e^{-\alpha t}dt\right]=\mathbb{E}\left[\int_0^{I+B}(r_1(\tau_1+M_1^A(t))+r_1'(\tau_1+M_1^A(t))h+O(h^2)e^{-\alpha t}dt\right]$$

Therefore,

$$\implies \frac{\mathbb{E}\Big[\int_0^{I+B}(r_1(\tau_1+h+M_1^A(t))-r_1(\tau_1+M_1^A(t)))e^{-\alpha t}\Big]}{h}dt=\mathbb{E}\left[\int_0^{I+B}(r_1'(\tau_1+M_1^A(t))+O(h))e^{-\alpha t}dt\right]$$

Therefore taking $h\to 0$ yields the desired derivative.

Let $f(\alpha,\tau_1,\tau_2)=\min_A \text{hold}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2,A)$. Then using [3, Theorem 1], we know

$$\frac{\partial}{\partial\tau_1}f(\alpha,\tau_1,\tau_2)=\frac{\partial}{\partial\tau_1}\text{hold}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2,A)\Big|_{A^*(\alpha,\tau_1,\tau_2)}$$

Suppose U_1,U_2 are optimal in CostMDP(α), we know $U_1,U_2\restriction\tau_1,\tau_2=\arg\min_A \text{hold}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2,A)$ by Lemma 6.16. Therefore

$$\frac{\partial}{\partial\tau_1}\text{hold}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2,U_1,U_2)=\frac{\mathbb{E}\Big[\int_0^{I+B}r_1'(\tau_1+M_1^{U_1,U_2\restriction\tau_1,\tau_2}(t))e^{-\alpha t}dt\Big]}{\frac{1}{\alpha}(1-\mathbb{E}[e^{-\alpha(I+B)}])}$$

Now taking $\alpha\to 0$, we have

$$\begin{aligned}\lim_{\alpha\to0}\frac{\partial}{\partial\tau_1}\text{hold}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2,U_1,U_2)&=\lim_{\alpha\to0}\frac{\mathbb{E}\Big[\int_0^{I+B}r_1'(\tau_1+M_1^{U_1,U_2\restriction\tau_1,\tau_2}(t))e^{-\alpha t}dt\Big]}{\frac{1}{\alpha}(1-\mathbb{E}[e^{-\alpha(I+B)}])}\\&=\frac{\mathbb{E}\Big[\int_0^{I+B}r_1'(\tau_1+M_1^{U_1,U_2\restriction\tau_1,\tau_2}(t))dt\Big]}{\mathbb{E}[I+B]}=\lambda_1\mathbb{E}[c_1(\tau_1+T_1^{U_1,U_2\restriction\tau_1,\tau_2})].\end{aligned}$$

The last equality follows similar to Lemma 6.19: if instantaneous cost in state (t_1,t_2) is $r_1'(t_1)$, we know by [1, Theorem 1] this is equivalent to the instantaneous cost in a system where each class 1 job with age t incurs $c_1'(t)$ and each class 2 job incurs 0 cost. Then the time average cost $\frac{\mathbb{E}\Big[\int_0^{I+B}r_1'(\tau_1+M_1^{U_1,U_2\restriction\tau_1,\tau_2}(t))dt\Big]}{\mathbb{E}[I+B]}$ equals the average class 1 arrival rate λ_1 times the average cost obtained by a class 1 job $\mathbb{E}[c_1(\tau_1+T_1^A)]$. □

Definition 6.22 (BP(t_1,t_2))**.** We define BP(t_1,t_2) to be the length of a 2 class M/M/1 busy period starting with the oldest job of class i being t_i. We must have $t_i\geq 0$ or $t_i\sim -I_i$. If $t_i\geq 0$, we have $1+\text{Pois}(\lambda_i t_i)$ jobs of class i present at $t=0$. If $t_i\sim I_i$, we start with no class i jobs, but Poisson arrivals in the future.

Proof of Lemma 6.11. We first show the case when $t_2>\tau_2$. Let $\delta\in(0,t_1-\tau_1)$. The scheduling policies defined by (τ_1,τ_2,U_1,U_2) and $(\tau_1+\delta,\tau_2,U_1,U_2)$ are both work conserving with respect to class 1 jobs older than $\tau_1+\delta$ and class 2 jobs older than $U_2^{-1}(U_1(\tau_1+\delta))$. Call these jobs "important" (see Figure 5). In state (t_1,t_2), we must be serving an important job. Let B'_δ be the end of a busy period with respect to this important queue. Namely, $B'_\delta=\min\{T: S_1(T)<\tau_1+\delta, S_2(T)<U_2^{-1}(U_1(\tau_1+\delta))\}$ starting from

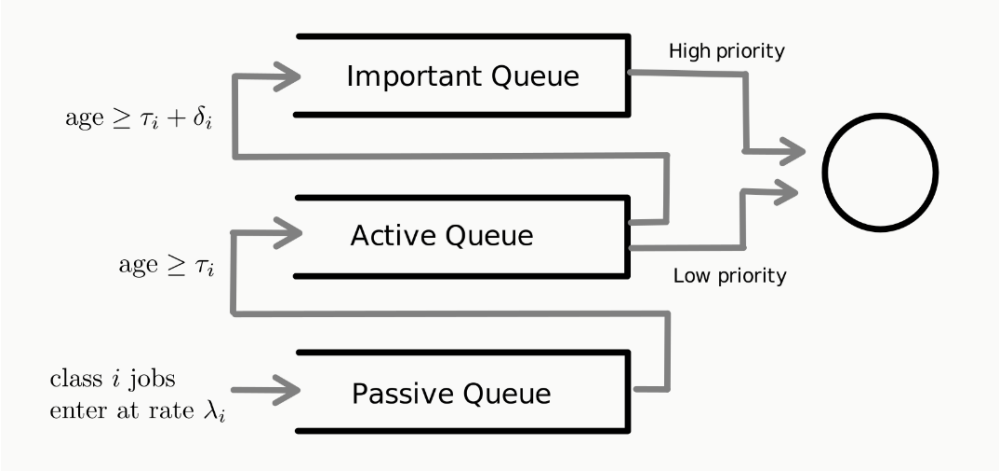


Figure 5: Active and important jobs under the scheduling policy $(\tau_1, \tau_2, U_1, U_2)$.

$S(0) = (t_1, t_2)$. Let $\delta_1 = \delta, \delta_2 = U_2^{-1}(U_1(\tau_1 + \delta)) - \tau_2$. Then $B'_\delta \cong \text{BP}(t_1 - \tau_1 - \delta_1, t_2 - \tau_2 - \delta_2)$. From state t_1, t_2, until time B'_δ, both $(\tau_1, \tau_2, U_1, U_2)$ and $(\tau_1 + \delta, \tau_2, U_1, U_2)$ incur identical costs and undergo identical transitions as determined by U_1, U_2. Thus we have[9]

$$\begin{aligned}
\text{hold}(\alpha, (t_1, t_2), \tau_1 + \delta, \tau_2, U_1, U_2) &= \int_0^{B'_\delta} \alpha \sum r_i(S_i^{U_1,U_2}(t)) e^{-\alpha t} dt \\
&\quad + e^{-\alpha B'_\delta} \text{hold}(\alpha, \tau_1 + \delta_1 - I_1, \tau_2 + \delta_2 - I_2, \tau_1 + \delta, \tau_2, U_1, U_2) \\
\text{hold}(\alpha, (t_1, t_2), \tau_1, \tau_2, U_1, U_2) &= \int_0^{B'_\delta} \alpha \sum r_i(S_i^{U_1,U_2}(t)) e^{-\alpha t} dt \\
&\quad + e^{-\alpha B'_\delta} \text{hold}(\alpha, \tau_1 + \delta_1 - I_1, \tau_2 + \delta_2 - I_2, \tau_1, \tau_2, U_1, U_2)
\end{aligned}$$

if $S_i^{U_1,U_2}(t)$ represents class i state at time t starting from $S(0) = (t_1, t_2)$, with transitions determined by

[9] all equations below are in expectation over long-run state transitions

U_1, U_2. Therefore

$$\frac{\partial}{\partial\tau_1}\operatorname{hold}(\alpha, S_0, \tau_1, \tau_2, U_1, U_2) = \lim_{\delta\to 0}\frac{1}{\delta}(\operatorname{hold}(\alpha,(t_1,t_2),\tau_1+\delta,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,(t_1,t_2),\tau_1,\tau_2,U_1,U_2))$$

$$= \lim_{\delta\to 0}\frac{e^{-\alpha B'_\delta}}{\delta}(\operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2+\delta_2-I_2,\tau_1+\delta,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2+\delta_2-I_2,\tau_1,\tau_2,U_1,U_2))$$

$$= \lim_{\delta\to 0}\frac{e^{-\alpha B'_\delta}}{\delta}(1-e^{\lambda_2\delta_2})(\operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2,\tau_1+\delta,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2,\tau_1,\tau_2,U_1,U_2))$$

$$+ \frac{e^{-\alpha B'_\delta}}{\delta}e^{\lambda_2\delta_2}(\operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2-I_2,\tau_1+\delta,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2))$$

(Casing on the event $I_2 > \delta_2$)

$$= e^{-\alpha B'}\lim_{\delta\to 0}\frac{1}{\delta}(\operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2-I_2,\tau_1+\delta,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,\tau_1+\delta-I_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2))$$

(First summand had limit 0 as $\delta\to 0$)

$$= e^{-\alpha B'}\lim_{\delta\to 0}\frac{1}{\delta}\Big(\operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2-I_2,\tau_1+\delta,\tau_2,U_1,U_2)$$

$$- e^{-\lambda_1\delta}\operatorname{hold}(\alpha,\tau_1-I_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2) - (1-e^{-\lambda_1\delta})\operatorname{hold}(\alpha,\tau_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2)\Big)$$

(Casing on the event $I_1 > \delta_1$)

$$= e^{-\alpha B'}\lim_{\delta\to 0}\frac{1}{\delta}(\operatorname{hold}(\alpha,\tau_1+\delta_1-I_1,\tau_2-I_2,\tau_1+\delta,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,\tau_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2))$$

$$+ e^{-\alpha B'}\lim_{\delta\to 0}\frac{1}{\delta}(1-e^{-\lambda_1\delta})(\operatorname{hold}(\alpha,\tau_1-I_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,\tau_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2))$$

$$= e^{-\alpha B'}\frac{\partial}{\partial\tau_1}\operatorname{hold}(\alpha,\tau_1-I_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2)$$

$$- \lambda_1 e^{-\alpha B'}(\operatorname{hold}(\alpha,\tau_1-I_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,\tau_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2))$$

We take the limit $\alpha\to 0$. Using Lemma 6.21, we can evaluate the limit of the first summand. By Lemma 6.19, we know the limit of the second summand is 0.

$$\lim_{\alpha\to 0}\frac{\partial}{\partial\tau_1}\operatorname{hold}(\alpha,S_0,\tau_1,\tau_2,U_1,U_2) = \lim_{\alpha\to 0}e^{-\alpha B'}\frac{\partial}{\partial\tau_1}\operatorname{hold}(\alpha,\tau_1-I_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2)$$
$$- \lambda_1\lim_{\alpha\to 0}e^{-\alpha B'}(\operatorname{hold}(\alpha,\tau_1-I_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,\tau_1,\tau_2-I_2,\tau_1,\tau_2,U_1,U_2))$$
$$= \lambda_1\mathbb{E}[c_1(\tau_1 + T_1^{U_1,U_2\restriction\tau_1,\tau_2})].$$

Now if $t_2 = \tau_2 - I_2$,

$$\begin{aligned}\operatorname{hold}(\alpha,(t_1,t_2),\tau_1+\delta,\tau_2,U_1,U_2) &= \alpha(r_1(t_1)+r_2(t_2))\delta_2 + o(\delta)\\ &\quad + \mu\delta_2\operatorname{hold}(\alpha,(t_1+\delta_2-I_2,\tau_2+\delta_2-I_2),\tau_1+\delta,\tau_2,U_1,U_2)\\ &\quad + e^{-\alpha\delta_2}(1-\mu\delta_2)\operatorname{hold}(\alpha,(t_1+\delta_2,\tau_2+\delta_2-I_2),\tau_1+\delta,\tau_2,U_1,U_2)\\ \operatorname{hold}(\alpha,(t_1,t_2),\tau_1,\tau_2,U_1,U_2) &= \alpha(r_1(t_1)+r_2(t_2))\delta_2 + o(\delta)\\ &\quad + \mu\delta_2\operatorname{hold}(\alpha,(t_1+\delta_2-I_2,\tau_2+\delta_2-I_2),\tau_1,\tau_2,U_1,U_2)\\ &\quad + e^{-\alpha\delta_2}(1-\mu\delta_2)\operatorname{hold}(\alpha,(t_1+\delta_2,\tau_2+\delta_2-I_2),\tau_1,\tau_2,U_1,U_2)\end{aligned}$$

(Lemmas 6.19 and 6.21)

$$\frac{\partial}{\partial\tau_1}\operatorname{hold}(\alpha,(t_1,t_2),\tau_1,\tau_2,U_1,U_2) = \lim_{\delta\to 0}\frac{1}{\delta}(\operatorname{hold}(\alpha,(t_1,t_2),\tau_1+\delta,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,(t_1,t_2),\tau_1,\tau_2,U_1,U_2))$$

$$= \lim_{\delta\to 0}\frac{e^{-\alpha\delta_2}}{\delta}(\operatorname{hold}(\alpha,(t_1+\delta_2,\tau_2+\delta_2-I_2),\tau_1+\delta,\tau_2,U_1,U_2) - \operatorname{hold}(\alpha,(t_1+\delta_2,\tau_2+\delta_2-I_2),\tau_1,\tau_2,U_1,U_2))$$

From initial state $\tau_2+\delta_2-I_2$, we are able to follow the same argument as the previous case ($t_2 > \tau_2$). From the starting state $(t_1, \tau_1+\delta-I_2)$, until time B'_δ, both (τ_1,τ_2,U_1,U_2) and $(\tau_1+\delta,\tau_2,U_1,U_2)$ incur identical

costs and undergo identical transitions as determined by U_1, U_2. [10] Thus we have here as well that

$$\lim_{\alpha\to 0}\frac{\partial}{\partial\tau_1}\operatorname{hold}(\alpha,(t_1,t_2),\tau_1,\tau_2,U_1,U_2)=\lambda_1\mathbb{E}[c_1(\tau_1+T_1^{U_1,U_2\restriction\tau_1,\tau_2})].$$

□

6.3.3 Service-time derivative

Proof of Lemma 6.12. First starting from the start of an active queue idle period (τ_1-I_1,τ_2-I_2), we know the server is idle for I time, then active for B time, and then returns to the starting state distribution. Thus

$$\begin{aligned}\operatorname{serve}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2)&=e^{-\alpha I}\int_0^B e^{-\alpha t}dt+e^{-\alpha(I+B)}\operatorname{serve}(\alpha,(\tau_1-I_1,\tau_2-I_2),\tau_1,\tau_2)\\&=\frac{e^{-\alpha I}(1-e^{-\alpha B})}{\alpha(1-e^{-\alpha(I+B)})}\end{aligned}$$

Let $\rho_\alpha=\frac{e^{-\alpha I}(1-e^{-\alpha B})}{\alpha(1-e^{-\alpha(I+B)})}$.

Now from state (t_1,t_2), under policy (τ_1,τ_2)[11], we first serve jobs for $B'=\mathrm{BP}(t_1-\tau_1,t_2-\tau_2)$ time, then we enter state (τ_1-I_1,τ_2-I_2). Thus we have

$$\operatorname{serve}(\alpha,(t_1,t_2),\tau_1,\tau_2)=\int_0^{B'}e^{-\alpha t}dt+e^{-\alpha B'}\rho_\alpha.$$

Similarly, from state (t_1,t_2), under policy $(\tau_1+\delta,\tau_2)$, we first serve jobs for $B'=\mathrm{BP}(t_1-\tau_1-\delta,t_2-\tau_2)$ time. We have

$$\operatorname{serve}(\alpha,(t_1,t_2),\tau_1+\delta,\tau_2)=\int_0^{B''}e^{-\alpha t}dt+e^{-\alpha B''}\rho_\alpha.$$

Here note $B'=B''+B_\delta$ where $B_\delta\cong\mathrm{BP}(\delta-I_1,-I_2)$. Thus

$$\begin{aligned}\frac{\partial}{\partial\tau_1}\operatorname{serve}(\alpha,(t_1,t_2),(\tau_1,\tau_2))&=\lim_{\delta\to0}\frac{1}{\delta}(\operatorname{serve}(\alpha,(t_1,t_2),\tau_1+\delta,\tau_2)-\operatorname{serve}(\alpha,(t_1,t_2),\tau_1\tau_2))\\&=\lim_{\delta\to0}\frac{1}{\delta}\left(\int_{B'}^{B''}e^{-\alpha t}dt+\rho_\alpha(e^{-\alpha B''}-e^{-\alpha B'})\right)\\&=\lim_{\delta\to0}\frac{-1}{\delta}\left(\frac{e^{-\alpha B''}-e^{-\alpha B'}}{\alpha}\right)(1-\alpha\rho_\alpha)\\&=-(1-\alpha\rho_\alpha)\lim_{\delta\to0}\frac{\mathbb{E}[e^{-\alpha B''}]}{\delta}\left(\frac{1-e^{-\alpha B_\delta}}{\alpha}\right)\\&=-(1-\alpha\rho_\alpha)\mathbb{E}[e^{-\alpha B'}]\lim_{\delta\to0}\frac{1}{\mathbb{E}[e^{-\alpha B_\delta}]}\left(\frac{1-\mathbb{E}[e^{-\alpha B_\delta}]}{\alpha\delta}\right)\end{aligned}$$

Recall $B_\delta\cong\mathrm{BP}(\delta-I_1,-I_2)$. Essentially B_δ is the busy period started by $N\sim\mathrm{Pois}(\lambda_1\delta)$ class 1 jobs. We know $\Pr[N=0]=e^{-\lambda_1\delta},\Pr[N=1]=\lambda_1\delta+o(\delta),\Pr[N\geq2]=o(\delta)$. Thus by casing on the value of N, we have

$$\mathbb{E}[e^{-\alpha B_\delta}]=1-\lambda_1\delta+\lambda_1\delta\mathbb{E}[e^{-\alpha B}]+o(\delta).$$

Therefore

$$\begin{aligned}\frac{\partial}{\partial\tau_1}\operatorname{serve}(\alpha,(t_1,t_2),(\tau_1,\tau_2))&=-(1-\alpha\rho_\alpha)\mathbb{E}[e^{-\alpha B'}]\lim_{\delta\to0}\frac{1}{\mathbb{E}[e^{-\alpha B_\delta}]}\left(\frac{1-\big(1-\lambda_1\delta(1-\mathbb{E}[e^{-\alpha B}])+o(\delta)\big)}{\alpha\delta}\right)\\&=-(1-\alpha\rho_\alpha)\mathbb{E}[e^{-\alpha B'}]\lambda_1\left(\frac{1-\mathbb{E}[e^{-\alpha B}]}{\alpha}\right)\xrightarrow{\alpha\to0}-(1-\rho)\cdot\lambda_1\cdot\frac{1/\mu}{1-\rho}=-\rho_1.\end{aligned}$$

[10] Note here $B'_\delta=\mathrm{BP}(t_1-\tau_1-\delta,-I_2)$ refers to a busy period starting with no class 2 jobs.
[11] note that the active queue scheduling policy does not affect service cost

The last limit can be seen as follows. First, $\lim_{\alpha\to 0} \alpha\rho_\alpha = \lim_{\alpha\to 0} \frac{e^{-\alpha I}(1-e^{-\alpha B})}{1-e^{-\alpha(I+B)}} = \frac{\mathbb{E}[B]}{\mathbb{E}[I+B]}$. This can be seen both using L'Hopital's rule and via the argument that $\alpha\rho_\alpha = \alpha\,\text{serve}(\alpha, (-I_1, -I_2), 0, 0)$ approaches the time average fraction of time spent in service as $\alpha \to 0$. Thus the first product term $1 - \alpha\rho_\alpha \to 1 - \rho$ as $\alpha \to 0$. Next clearly $\lim_{\alpha\to 0} \mathbb{E}[e^{-\alpha B'}] = 1$. Lastly $\frac{1-\mathbb{E}[e^{-\alpha B}]}{\alpha} \to \mathbb{E}[B]$ as $\alpha \to 0$. We know $\mathbb{E}[B] = \frac{1/\mu}{1-\rho}$. Thus we know the limit of each term in the above product as $\alpha \to 0$. The limit of the product equals the product of the limits. □

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